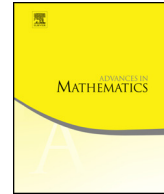




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The tropical Nullstellensatz for congruences



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ABSTRACT

We develop the foundations of a theory of algebraic geometry for semirings, concentrating mainly on the semiring of tropical polynomials. Replacing ideals with the more general notion of congruences, we establish a relationship between congruences on the semiring of tropical polynomials and subsets of tropical space. We ultimately establish analogues of the weak and strong forms of Hilbert's Nullstellensatz.

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1. Introduction

1.1. Tropical geometry

Tropical algebraic geometry originally developed as the study of certain geometric objects, called *amoebas*, that arose as the logarithmic images of complex varieties. These amoebas possessed semi-linear “skeletons” in which was encoded much of the geometric

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content of the original complex variety. The skeletons could also be recovered as limits of logarithmic (or valuative) maps from the original complex varieties, and soon the skeletons came to be called *tropical varieties*. For a comprehensive overview of the field, see [49,17,51,27,10], or [43].

A unifying characteristic of tropical varieties is their inherently piecewise-linear nature. One benefit of this linearity has been their ability to translate algebro-geometric questions into questions of combinatorics, graph theory and linear algebra. As only a small sampling, tropical algebraic geometry has been used to: compute enumerative invariants of real and complex projective spaces [8,9]; give new proofs of the classical Brill–Noether theorem [15] and the Gieseker–Petri theorem [32]; and provide new approaches to the maximal rank conjecture [33], the Hodge conjecture [3], Rota’s conjecture [25], and the uniformity conjecture for rational points on curves [37].

As the field has developed, the number of algebro-geometric objects with tropical analogues has multiplied. For example, there are now tropical analogues of: the Grassmannian [50] and the Grassmann algebra [21]; the Riemann–Roch theorem [4,18]; the Torelli theorem [7,11,13]; Hurwitz numbers [12] and genus 0 characteristic numbers [6]; Hodge theory [47,2]; and homology [28]. There has also been a great deal of work towards understanding tropicalization and the connection between classical and tropical structures, e.g., [20,1] and [5].

Concurrent with the above developments, there has been an explosion in the number of programs attempting to establish a foundational theory for the subject, with a diverse variety of approaches. There is the toric approach of Mikhalkin and others, involving polyhedra and balancing conditions, e.g., [46]. Many are exploring the intimate connection between tropical varieties, Berkovich spaces and $\mathbf{F}_1$ -geometry (e.g., [52]); closely related are the congruence schemes of Deitmar [16]. Macpherson is developing a theory of skeleta [44], which provides a unified framework for the study of both tropical varieties and skeleta of non-Archimedean analytic spaces. Others are pursuing the combinatorial aspects of tropical objects, e.g., [26,14]. There is the theory of layered categories pioneered by Izhakian et al. [30,31], and also the development of algebraic geometry over the new hyperring and hyperfield structures of Viro and Jun; see [53,55,54,35,36]. More closely related to the paper at hand are the tropical schemes of Giansiracusa–Giansiracusa [19] and MacLagan–Rincon [42]. And recently several of the aforementioned programs may have found their apotheosis in the ordered blueprint schemes of Lorscheid; see [38,40,39,41].

The current paper is similar in spirit to the above programs, in that it aims to construct a general framework for algebraic geometry in the tropical setting. We begin with the goal of understanding geometric subsets of tropical n -space at which a collection of pairs of tropical polynomials agree; e.g., subsets of the form $\{\mathbf{a} \mid f(\mathbf{a}) = g(\mathbf{a})\}$ for some tropical polynomials f and g . In the classical setting such a locus is equivalent to the vanishing locus of the ideal $\langle f - g \rangle$. Since subtraction does not exist for tropical polynomials, we instead deal directly with the pair (f, g) (or, in general, a collection of such pairs). In order to have a coherent theory, we are led to consider algebraic objects on semirings that

behave both like ideals and equivalence relations. Fortunately such a structure already exists, in the form of a *congruence* on a semiring. The focus of the current paper is an algebro-geometric study of the properties of congruences and their associated congruence varieties, with the goal of obtaining an analogue of Hilbert’s Nullstellensatz.

We note that our goal is not to formalize the geometry arising from the tropicalization of classical algebro-geometric objects. Within this paper, the objects constructed are studied for their intrinsic properties, and not in how they connect to standard tropical varieties. That being said, tropical varieties do fit into the current framework (see [Remark 3.3](#)), and interesting future work will involve exploring this inclusion in greater detail.

It should also be noted that this is not the first search for an analogue of Hilbert’s Nullstellensatz in the tropical setting. An analogue in the case of tropical vanishing loci (which are distinct from the congruence varieties studied here) was found in [\[48\]](#) and [\[29\]](#). As for congruences, there is the recent work of Grigoriev–Podolskii [\[24\]](#) and Joó–Mincheva [\[34\]](#). The former focuses on tropical polynomials and includes, among other results, a version of the tropical analogue of the weak form of the Nullstellensatz ([Theorem 2](#) below; [\[24, Theorem 8\]](#)). The work of Joó–Mincheva is even more closely related to the current paper, especially in its strong focus on congruences. The central feature of [\[34\]](#), however, is a notion of primality for congruences defined in terms of the twisted product (defined below), whereas in the current paper prime congruences are only lightly touched on. Using completely different methods, they prove a strengthened version of [Theorem 5](#); see [\[34, Theorem 5.3\]](#), as well as [Remark 5.6](#).

1.2. Overview

The paper is structured as follows. [Section 2](#) is devoted to establishing the algebraic foundation on which we will be working, focusing especially on congruences. A *congruence* on a semiring R is a subsemiring of $R \times R$ that also defines an equivalence relation on R . The bulk of [Section 2](#) is devoted to establishing their basic properties, such as induced quotient semirings and prime congruences. In [Section 3](#) we focus our attention on $\mathbf{T}[\mathbf{x}]$, the semiring of tropical polynomials. We establish the fundamental correspondence between congruences on $\mathbf{T}[\mathbf{x}]$ and subsets of $\mathbf{T}^n$: to each congruence E on $\mathbf{T}[\mathbf{x}]$ we associate its *congruence variety*, $\mathbf{V}(E) \subseteq \mathbf{T}^n$, consisting of those points $\mathbf{a}$ for which $f(\mathbf{a}) = g(\mathbf{a})$ for every $(f, g) \in E$; and conversely, to any subset $S \subseteq \mathbf{T}^n$ we associate a congruence, $\mathbf{E}(S)$, consisting of all pairs (f, g) for which $f(\mathbf{a}) = g(\mathbf{a})$ for every $\mathbf{a} \in S$. As in classical algebraic geometry, two questions naturally arise. First, when is the set $\mathbf{V}(E)$ nonempty? And secondly, what is the algebraic structure of $\mathbf{E}(\mathbf{V}(E))$? Answers to these questions represent analogues of the weak and strong forms of Hilbert’s Nullstellensatz, respectively. We lay the foundations for those results by first analyzing the case when $\mathbf{V}(f, g)$ is empty, and also the case when $\mathbf{V}(f, g) = \mathbf{T}^n$. The analysis of the latter involves an auxiliary notion of “saturation” for tropical polynomials, a property which we use to algebrize the condition that two tropical polynomials define the same tropical function.

This ultimately leads to the following description of $\mathbf{E}(\mathbf{V}(E))$ in the special case of the trivial congruence:

Theorem 1. *The congruence $\mathbf{E}(\mathbf{T}^n)$ is the intersection of all prime congruences on $\mathbf{T}[\mathbf{x}]$.*

Section 4 is devoted to answering the question of when $\mathbf{V}(E)$ is empty. The analysis proceeds by a reduction to the principally generated case, together with an auxiliary notion of “flatness” for congruences on $\mathbf{T}[\mathbf{x}]$. The result is the following weak form of the tropical Nullstellensatz for congruences:

Theorem 2. *Suppose E is a finitely generated congruence on $\mathbf{T}[\mathbf{x}]$. Then $\mathbf{V}(E)$ is empty if and only if there exists $h \in \mathbf{T}[\mathbf{x}]$ with nonzero constant term such that $(h, \epsilon h) \in E$ for some $\epsilon > 0$ (equivalently, all $\epsilon \in \mathbf{R}$).*

The remainder of the paper is devoted to understanding the algebraic structure of $\mathbf{E}(\mathbf{V}(E))$. The first issue is to find a candidate for $\mathbf{E}(\mathbf{V}(E))$, which in the classical case is the radical ideal. It turns out to be a slightly more complicated object in the tropical setting. First, given a congruence E on $\mathbf{T}[\mathbf{x}]$ we define a set $\text{rad}^-(E)$ called the *pre-radical* of E , consisting of all pairs (f, g) for which the following condition is satisfied: there exist $r \in \mathbf{T}[\mathbf{x}]$, $\epsilon \neq 1_{\mathbf{T}}$ and $N > 0$ such that

$$(1_{\mathbf{T}}, \epsilon) \rtimes ((f + g)^N + r, 0_{\mathbf{T}}) \rtimes (f, g) \in E.$$

Here we are using the so-called *twisted product* (introduced in Section 2), defined by $(a, b) \rtimes (c, d) = (ac + bd, ad + bc)$. The association $E \mapsto \mathbf{V}(E)$ behaves well with respect to twisted products, in that $\mathbf{V}((h, k) \rtimes (f, g)) = \mathbf{V}(h, k) \cup \mathbf{V}(f, g)$. This property, together with Theorem 2 and a few observations about pairs of the form $(f, g)^n$ and $(f, g)^{\rtimes n}$, naturally lead one to consider the set defined above. One always has $E \subseteq \text{rad}^-(E) \subseteq \mathbf{E}(\mathbf{V}(E))$ and $\mathbf{V}(E) = \mathbf{V}(\text{rad}^-(E))$. Moreover, with this new notation Theorem 2 can be restated as follows:

Theorem 3. *If E is a finitely generated congruence on $\mathbf{T}[\mathbf{x}]$, then $\mathbf{V}(E)$ is empty if and only if $(1_{\mathbf{T}}, 0_{\mathbf{T}}) \in \text{rad}^-(E)$.*

We begin our final analysis of $\mathbf{E}(\mathbf{V}(E))$ by considering the special case of a pair $(f, 0_{\mathbf{T}})$. We pursue a strategy analogous to the so-called “Rabinowitsch trick,” attempting to “invert” the function f by adjoining the pair $(1_{\mathbf{T}}, yf)$ to the congruence E , working now on the semiring $\mathbf{T}[\mathbf{x}, y]$. The resulting congruence on $\mathbf{T}[\mathbf{x}, y]$ has empty associated congruence variety, and so Theorem 3 applies. After “clearing denominators,” we arrive at the following result:

Theorem 4. *If E is a finitely generated congruence on $\mathbf{T}[\mathbf{x}]$, then $(f, 0_{\mathbf{T}}) \in \mathbf{E}(\mathbf{V}(E))$ if and only if $(f, 0_{\mathbf{T}}) \in \text{rad}^-(E)$.*

The general case is more complicated. First, transitivity considerations force us to consider a slight extension of $\text{rad}^-(E)$ to a set we call the *radical* of E ; see Section 5. Secondly, there is an additional phenomenon that arises in the tropical setting not seen in the classical setting. By way of example, suppose one knew $(f + t, t) \in \mathbf{E}(\mathbf{V}(E))$ for every $t > 1_{\mathbf{T}}$, i.e., that $f(\mathbf{a}) \leq t$ for every $\mathbf{a} \in \mathbf{V}(E)$. By continuity one would then have $f(\mathbf{a}) \leq 1_{\mathbf{T}}$, and hence $(f + 1_{\mathbf{T}}, 1_{\mathbf{T}}) \in \mathbf{E}(\mathbf{V}(E))$. In other words, the congruence $\mathbf{E}(\mathbf{V}(E))$ is somehow “closed under limits,” a property not shared by congruences in general. With this example in mind, we proceed to define a set $(\text{rad}(E))^{\text{lim}}$, consisting of all *limit pairs* of *ascending sequences* in $\text{rad}(E)$. This set might not be transitively closed, so we let $\hat{E}$ denote its “one-link transitive closure.” (See Section 5 for complete details.) We are ultimately able to prove the following strong form of the tropical Nullstellensatz for congruences:

Theorem 5. *If E is a finitely generated congruence on $\mathbf{T}[\mathbf{x}]$, then $\mathbf{E}(\mathbf{V}(E)) = \hat{E}$.*

The proof proceeds by reducing the general case of $(f, g) \in \mathbf{E}(\mathbf{V}(E))$ to the special case when $g = 1_{\mathbf{T}}$, using a second application of the Rabinowitsch trick to “invert” the function g . We then use Theorem 3 to prove $(f + t, t) \in \text{rad}(E)$ and $(f + t^{-1}, f) \in \text{rad}(E)$ for every $t > 1_{\mathbf{T}}$. The result follows.

1.3. Acknowledgments

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2. Semirings and congruences

2.1. Semirings

The main algebraic structures we will be dealing with are semirings. Recall that a **semiring** $(R, +, \cdot)$ is a nonempty set R together with two binary operations, satisfying the following:

- (i) $(R, +)$ is a commutative monoid with identity element 0;
- (ii) $(R, \cdot)$ is a monoid with identity element $1 \neq 0$;
- (iii) $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$ for all $a, b, c \in R$;
- (iv) $0 \cdot a = 0 = a \cdot 0$ for all $a \in R$.

A **morphism** $f : R \rightarrow S$ of semirings is set map compatible with the semiring structures, i.e., compatible with the operations and mapping 0_R to 0_S and 1_R to 1_S . A semiring R is **commutative** if the monoid $(R, \cdot)$ is commutative. All semirings we consider will be commutative. A **subsemiring** of a semiring R is a subset that inherits the structure of a semiring from R .

Example 2.1. The set $\mathbf{B} = \{0, 1\}$ is a semiring under the usual addition and multiplication, with the notable exception that $1 + 1 = 1$. This is usually called the **boolean semiring**; see [22, pg. 12].

A semiring R is **additively idempotent** if the addition operation is idempotent, i.e., if $a + a = a$ for every $a \in R$. Additively idempotent semirings are precisely $\mathbf{B}$ -semialgebras: the only nontrivial requirement for $\mathbf{B}$ to act on a semiring R is that $a = 1_{\mathbf{B}} \cdot a = (1_{\mathbf{B}} + 1_{\mathbf{B}}) \cdot a = a + a$ for all $a \in R$. The semiring $\mathbf{B}$ is the initial object in the category of additively idempotent semirings. All semirings we consider will be additively idempotent.

Example 2.2. The set $\mathbf{N} \cup \{\infty\}$ is an additively idempotent semiring under the operations of minimum and addition, where minimum plays the role of the additive operation and addition the role of the multiplicative operation. The additive identity is ∞ , while the multiplicative identity is the natural number 0. This is a subsemiring of $(\mathbf{R} \cup \{\infty\}, \min, +)$, the so-called **optimization algebra**; see [22, pp. 15–16]. Along similar lines, the set $\mathbf{N} \cup \{-\infty\}$ is an additively idempotent semiring under the operations of maximum and addition. It is a subsemiring of $(\mathbf{R} \cup \{-\infty\}, \max, +)$, which we will call the **tropical semiring** and denote by $\mathbf{T}$. Note that the additive identity $0_{\mathbf{T}}$ is the element $-\infty$, and the multiplicative identity $1_{\mathbf{T}}$ is the real number 0. (In some texts, the semiring $(\mathbf{R} \cup \{-\infty\}, \max, +)$ is called the **schedule algebra** or **(max, +)-algebra**; the term tropical semiring is then reserved for $(\mathbf{N} \cup \{\infty\}, \min, +)$.) All semirings we focus on will be $\mathbf{T}$ -semialgebras.

Remark 2.3. For consistency and clarity of notation, we will use throughout the standard “+” and “ $\cdot$ ” notation for addition and multiplication in semirings, even in the case of calculations involving elements of $\mathbf{T}$. Unless explicitly stated otherwise, all calculations and formulas involving tropical numbers should be assumed to be “tropical.” For example, the tropical product $2 \cdot 3$ evaluates in classical operations as $2 + 3 = 5$, and the tropical function $f(x) = x^2 + x + 3$ evaluates in classical operations as $f(x) = \max\{2x, x, 3\}$. In those few situations in which it is decidedly more convenient to use the classical operations, we will always explicitly state we are doing so.

2.2. Congruences

We are ultimately interested in subsets of $\mathbf{T}^n$ on which collections of pairs of tropical polynomials agree; e.g., subsets of the form $\{\mathbf{a} \in \mathbf{T}^n \mid f(\mathbf{a}) = g(\mathbf{a})\}$ for some tropical

polynomials $f, g \in \mathbf{T}[\mathbf{x}]$. If we were working in a ring, such a locus would be equivalent to the vanishing locus of the ideal $\langle f - g \rangle$. When working in a semiring, however, we cannot reduce our original locus to the vanishing set of such an ideal. Instead, we must work directly with the original locus. This suggests that instead of studying ideals in R , we should instead study subsets $E \subseteq R \times R$ given by “relations.” Fortunately, such an algebraic structure already exists.

Definition 2.4. A **congruence** on a semiring R is a subset $E \subseteq R \times R$ with the following properties:

- (E1: Reflexivity) For every $a \in R$ we have $(a, a) \in E$;
- (E2: Symmetry) If $(a, b) \in E$, then $(b, a) \in E$;
- (E3: Transitivity) If $(a, b), (b, c) \in E$, then $(a, c) \in E$;
- (I1: Additive Closure) If $(a, b), (a', b') \in E$, then $(a + a', b + b') \in E$; and
- (I2: Multiplicative Closure) If $(a, b), (a', b') \in E$, then $(aa', bb') \in E$.

In other words, a congruence on R is a subsemiring of $R \times R$ that also defines an equivalence relation on R . It turns out that condition (I2) can be replaced with a condition similar to that in the definition of an ideal. For notational convenience, we first define a second product on $R \times R$.

Definition 2.5. Let R be a semiring. The **twisted product** on $R \times R$ is defined by

$$(a, b) \rtimes (c, d) := (ac + bd, ad + bc)$$

for $a, b, c, d \in R$.

The twisted product is the product one would expect to encounter if pairs (a, b) were represented as differences $a - b$. Indeed, observe that $(a - b)(c - d) = (ac + bd) - (ad + bc)$. This product is commutative and behaves well with respect to addition. Moreover, as mentioned above it can be used to give a slightly modified definition of a congruence.

Lemma 2.6. Suppose R is a commutative semiring. A subset $E \subseteq R \times R$ is a congruence on R if and only if E satisfies conditions (E1)–(E3), (I1), and the following condition:

(I2)' If $(a, b) \in E$, then $(c, d) \rtimes (a, b) \in E$ for every $c, d \in R$.

Proof. First suppose E is a congruence on R . For any $(a, b) \in E$ and $c, d \in R$ we have $(c, c), (d, d) \in E$ by (E1). Then by (I2) we have $(ca, cb), (da, db) \in E$. By properties (E2) and (I1), we then have that E contains $(c, d) \rtimes (a, b) = (ca + db, cb + da)$. Thus, E satisfies condition (I2)'.

Conversely, suppose E satisfies conditions (E1)–(E3), (I1), and (I2)'. For any $(a, b), (a', b') \in E$ we then have that E contains $(a', 0) \rtimes (a, b) = (a'a, a'b)$, and sim-

ilarly $(b, 0) \times (a', b') = (ba', bb')$. By condition (E3), it follows that $(aa', bb') \in E$. Thus, E satisfies condition (I2), and hence E is a congruence on R . $\square$

We will see later that twisted products behave well with respect to congruence varieties; see [Lemma 3.1](#).

Remark 2.7. This twisted product is the same as the one defined in [\[34, p. 4\]](#). However, note that in [\[34\]](#) the twisted product is the only product on pairs considered, and is simply denoted $(a, b)(c, d)$.

Remark 2.8. When working in a ring, there is an inclusion-preserving bijection between ideals and congruences, via the associations

$$\mathbf{E}(I) = \{(a, b) \mid a - b \in I\} \subseteq R \times R, \quad \mathbf{I}(E) = \{a - b \mid (a, b) \in E\} \subseteq R.$$

The **kernel** of a semiring morphism $f : R \rightarrow S$ is the (proper) congruence

$$\ker f = \{(a, b) \in R \times R \mid f(a) = f(b)\}.$$

2.3. Quotient semirings

If E is a congruence on a semiring R , then the set R/E of all equivalence classes induced by E is a semiring, with addition and multiplication defined in the usual way, i.e., by $[a] + [b] = [a + b]$ and $[a][b] = [ab]$. This semiring is called the **quotient** (or **factor**) **semiring** of R by E . There is a canonical surjective semiring morphism $\pi_E : R \rightarrow R/E$, given by $r \mapsto [r]$, with $\ker \pi_E = E$. By construction, there is a natural bijection between proper congruences on R and (isomorphism classes of) surjective semiring morphisms from R . Moreover, under this bijection containment of congruences corresponds to factorization of semiring morphisms. As a corollary, there is a natural inclusion-preserving bijection between congruences on a quotient semiring R/E and congruences on R that contain E .

Example 2.9. Define a map $\mathbf{T} \rightarrow \mathbf{B}$ by sending $0_{\mathbf{T}}$ to $0_{\mathbf{B}}$ and all other t to $1_{\mathbf{B}}$. This is a surjective semiring morphism; the key observation is that if $a, b \neq 0_{\mathbf{T}}$, then $a + b \neq 0_{\mathbf{T}} \neq ab$. (Note that these are tropical operations.) The corresponding congruence on $\mathbf{T}$ is

$$E_{\mathbf{B}} = (\mathbf{T} \times \mathbf{T}) \setminus \{(t_{\mathbf{T}}, 0_{\mathbf{T}}), (0_{\mathbf{T}}, t_{\mathbf{T}}) \mid t \neq 0_{\mathbf{T}}\}.$$

This congruence is maximal among proper congruences on $\mathbf{T}$: any congruence E properly containing it would contain a pair of the form $(t, 0_{\mathbf{T}})$ for some $t \neq 0_{\mathbf{T}}$. Every $t \neq 0_{\mathbf{T}}$ corresponds to a real number, and so has a tropical multiplicative inverse – its classical additive inverse. By property (I2)' the congruence E would then contain $(t^{-1}, 0_{\mathbf{T}}) \times (t, 0_{\mathbf{T}}) = (1_{\mathbf{T}}, 0_{\mathbf{T}})$ and hence be improper.

We claim $E_{\mathbf{B}}$ is the unique nontrivial proper congruence on $\mathbf{T}$. Suppose E is any nontrivial proper congruence on $\mathbf{T}$. As E is nontrivial, there exist some distinct $r, s \in \mathbf{T}$ with $(r, s) \in E$. As E is proper, we must also have $r, s \neq 0_{\mathbf{T}}$, i.e., r and s are real numbers. Without loss of generality, suppose $r < s$. Then E contains $(r^{-1}, 0_{\mathbf{T}}) \bowtie (r, s) = (1_{\mathbf{T}}, sr^{-1})$. It follows that $(1_{\mathbf{T}}, (sr^{-1})^n) \in E$ for every $n \in \mathbf{N}$. Similarly, $(1_{\mathbf{T}}, (rs^{-1})^n) \in E$ for every $n \in \mathbf{N}$. Now fix any $t \in \mathbf{T} \setminus \{0_{\mathbf{T}}\}$. Then t is a real number, so by the usual Archimedean property of $\mathbf{R}$ there exists some $n \in \mathbf{N}$ with $|t| < n(s - r)$ in classical operations; equivalently, $|t| < (sr^{-1})^n$ in tropical operations. If $t \geq 0 = 1_{\mathbf{T}}$, then observe that we have

$$(t, (sr^{-1})^n) = (1_{\mathbf{T}}, (sr^{-1})^n) + (t, t) \in E.$$

By transitivity, it follows that $(1_{\mathbf{T}}, t) \in E$. On the other hand, if $t \leq 1_{\mathbf{T}}$, then observe that we have

$$(1_{\mathbf{T}}, t) = (1_{\mathbf{T}}, (rs^{-1})^n) + (t, t) \in E.$$

We thus have $(1_{\mathbf{T}}, t) \in E$ for every $t \in \mathbf{T} \setminus \{0_{\mathbf{T}}\}$, and hence $E_{\mathbf{B}} \subseteq E$. As $E_{\mathbf{B}}$ is maximal among proper congruences, equality holds.

2.4. Generation, restriction and extension of congruences

Given any subset $\mathcal{S} \subseteq R \times R$, the **congruence generated by $\mathcal{S}$** is the smallest congruence on R that contains $\mathcal{S}$. We denote this congruence by $\langle \mathcal{S} \rangle$. If we wish to emphasize the semiring on which this is a congruence, we write $\langle \mathcal{S} \rangle_R$. Since an arbitrary intersection of congruences is a congruence, we can express $\langle \mathcal{S} \rangle$ as the intersection of all congruences on R that contain $\mathcal{S}$.

Example 2.10. Let R be any semiring and let $\text{ev}_0 : R[x] \rightarrow R$ be the evaluation-at-zero morphism. We claim $\ker \text{ev}_0 = \langle (x, 0) \rangle$. First notice that we certainly have $(x, 0) \in \ker \text{ev}_0$, and hence $\langle (x, 0) \rangle \subseteq \ker \text{ev}_0$. We next claim that $(f(x), f(0)) \in \langle (x, 0) \rangle$ for every $f \in R[x]$. Indeed, first observe that for every $a \in R$ and $d \geq 1$ we have $(ax^d, 0) = (ax^{d-1}, 0) \bowtie (x, 0) \in \langle (x, 0) \rangle$ by property (I2)'. Thus, if $f(x) = \sum_d a_d x^d$, then

$$(f(x), f(0)) = \left(\sum_d a_d x^d, a_0 \right) = (a_0, a_0) + \sum_{d \geq 1} (a_d x^d, 0) \in \langle (x, 0) \rangle,$$

by properties (E1) and (I1). From this it follows that if $f(0) = g(0)$, then by transitivity (of $(f(x), f(0)), (g(0), g(x)) \in \langle (x, 0) \rangle$) we have $(f(x), g(x)) \in \langle (x, 0) \rangle$, and hence $\ker \text{ev}_0 \subseteq \langle (x, 0) \rangle$.

Unlike the situation with ideals in rings, a general member of a finitely generated congruence cannot necessarily be expressed as a sum of multiples of the generators. The

transitivity property generally makes any description in terms of generators complicated. However, we can say the following:

Lemma 2.11. *Suppose R is a semiring and $\mathcal{S} \subseteq R \times R$ satisfies properties (E1), (E2), (I1), and (I2). Then $\langle \mathcal{S} \rangle$ consists of all pairs (f, g) for which there exists a finite transitive chain $(f, r_1), (r_1, r_2), \dots, (r_{n-1}, r_n), (r_n, g) \in \mathcal{S}$.*

Proof. Let E denote the set of all such pairs (f, g) as described above. We certainly have $\mathcal{S} \subseteq E$. We claim E is a congruence on R . Since $\mathcal{S}$ satisfies (E1) we have $\Delta \subseteq \mathcal{S} \subseteq E$, and so E also satisfies (E1). To see E satisfies (E2), suppose $(f, g) \in E$, so that $(f, r_1), \dots, (r_n, g) \in \mathcal{S}$ for some $r_i \in R$. As $\mathcal{S}$ satisfies (E2) we then have $(g, r_n), \dots, (r_1, f) \in \mathcal{S}$, and hence $(g, f) \in E$. To verify E satisfies (E3), suppose $(f, g), (g, h) \in E$, so that $(f, r_1), \dots, (r_n, g) \in \mathcal{S}$ and $(g, s_1), \dots, (s_m, h) \in \mathcal{S}$ for some $r_i, s_j \in R$. Concatenating these two lists gives a transitive chain $(f, r_1), \dots, (r_n, g), (g, s_1), \dots, (s_m, h) \in \mathcal{S}$, and hence $(f, h) \in E$. Finally, to verify E satisfies (I1) and (I2), suppose $(f_1, g_1), (f_2, g_2) \in E$, so that $(f_1, r_1), \dots, (r_n, g_1) \in \mathcal{S}$ and $(f_2, s_1), \dots, (s_m, g_2) \in \mathcal{S}$ for some $r_i, s_j \in R$. Inserting trivial pairs (i.e., pairs of the form (t, t)) if necessary, we may assume $m = n$. Since $\mathcal{S}$ satisfies (I1) and (I2), we then have that $(f_1 + f_2, r_1 + s_1), \dots, (r_n + s_n, g_1 + g_2) \in \mathcal{S}$ and $(f_1 f_2, r_1 s_1), \dots, (r_n s_n, g_1 g_2) \in \mathcal{S}$, and hence $(f_1 + f_2, g_1 + g_2) \in E$ and $(f_1 f_2, g_1 g_2) \in E$. Thus, E satisfies (I1) and (I2).

We have thus shown E is a congruence that contains $\mathcal{S}$. Now suppose E' is any congruence that contains $\mathcal{S}$. Observe that if we have a transitive chain $(f, r_1), \dots, (r_n, g) \in \mathcal{S} \subseteq E'$, then by transitivity in E' we must have $(f, g) \in E'$. Thus, $E \subseteq E'$. It follows that E is the smallest congruence on R that contains $\mathcal{S}$, and hence $E = \langle \mathcal{S} \rangle$. $\square$

Corollary 2.12. *Suppose E is a finitely generated congruence on a semiring R , say $E = \langle (a_1, b_1), \dots, (a_k, b_k) \rangle$. Let $\mathcal{S} \subseteq R \times R$ denote the collection of all pairs of the form*

$$\left(\sum_{\mathbf{m}, \mathbf{n} \in \mathbf{Z}_{\geq 0}^k} r_{\mathbf{m}, \mathbf{n}} \mathbf{a}^{\mathbf{m}} \mathbf{b}^{\mathbf{n}}, \sum_{\mathbf{m}, \mathbf{n} \in \mathbf{Z}_{\geq 0}^k} r_{\mathbf{m}, \mathbf{n}} \mathbf{a}^{\mathbf{n}} \mathbf{b}^{\mathbf{m}} \right),$$

where all but finitely many $r_{\mathbf{m}, \mathbf{n}} \in R$ are zero. Then E consists of all pairs (f, g) for which there exists a finite transitive chain $(f, r_1), (r_1, r_2), \dots, (r_{n-1}, r_n), (r_n, g) \in \mathcal{S}$.

Proof. Observe that the set $\mathcal{S}$ is the smallest subset of $R \times R$ that contains $(a_1, b_1), \dots, (a_k, b_k)$ and satisfies properties (E1), (E2), (I1), and (I2). Since E is the smallest subset with those properties that also satisfies (E3), it follows that $E = \langle \mathcal{S} \rangle$. The result then follows from the previous lemma. $\square$

Suppose $\phi : R \rightarrow S$ is a semiring morphism. Given any congruence E on R , one can ask whether the set $(\phi \times \phi)(E) = \{(\phi(e_1), \phi(e_2)) \mid (e_1, e_2) \in E\}$ is a congruence on S .

In order to contain Δ_S , one needs ϕ to be surjective. Even when ϕ is surjective, however, transitivity is still an issue. In fact, transitivity is the only issue:

Lemma 2.13. *Suppose $\phi : R \rightarrow S$ is a surjective semiring morphism and E is a congruence on R . Then the set $(\phi \times \phi)(E)$ satisfies properties (E1), (E2), (I1), and (I2).*

Proof. We verify each property in turn. Let $\mathcal{T} = (\phi \times \phi)(E)$. Since ϕ is surjective, $(\phi \times \phi)(\Delta_R) = \Delta_S$, and hence $\Delta_S \subseteq \mathcal{T}$. Thus $\mathcal{T}$ satisfies (E1). Property (E2) on $\mathcal{T}$ is immediate from property (E2) on E : if $(\phi(f), \phi(g)) \in \mathcal{T}$, where $(f, g) \in E$, then $(g, f) \in E$ and hence $(\phi(g), \phi(f)) \in \mathcal{T}$. Finally, suppose $(\phi(f_1), \phi(g_1)), (\phi(f_2), \phi(g_2)) \in \mathcal{T}$ for some $(f_1, g_1), (f_2, g_2) \in E$. The $(f_1 + f_2, g_1 + g_2), (f_1 f_2, g_1 g_2) \in E$, and hence (since ϕ is a semiring morphism) $(\phi(f_1) + \phi(f_2), \phi(g_1) + \phi(g_2)), (\phi(f_1)\phi(f_2), \phi(g_1)\phi(g_2)) \in \mathcal{T}$. Thus, $\mathcal{T}$ satisfies properties (I1) and (I2). $\square$

If $\phi : R \rightarrow S$ is a surjective semiring morphism and E is a congruence on R , define the **image congruence of E** to be the congruence $\langle (\phi \times \phi)(E) \rangle_S$, and denote it $\phi_*(E)$.

Corollary 2.14. *Suppose $\phi : R \rightarrow S$ is a surjective semiring morphism and E is a congruence on R . Then $\phi_*(E)$ consists of all pairs $(s_1, s_2) \in S \times S$ for which there exists a finite transitive chain $(s_1, t_1), (t_1, t_2), \dots, (t_{n-1}, t_n), (t_n, s_2) \in (\phi \times \phi)(E)$.*

Proof. This follows immediately from Lemmas 2.11 and 2.13. $\square$

Corollary 2.15. *Suppose $\phi : R \rightarrow S$ is a surjective semiring morphism and E is a congruence on R . If $E = \langle (a_1, b_1), \dots, (a_k, b_k) \rangle$, then*

$$\phi_*(E) = \langle (\phi(a_1), \phi(b_1)), \dots, (\phi(a_k), \phi(b_k)) \rangle.$$

Proof. Let $\mathcal{S}$ denote the collection of all pairs of the form

$$\left(\sum_{\mathbf{m}, \mathbf{n} \in \mathbf{Z}_{\geq 0}^k} r_{\mathbf{m}, \mathbf{n}} \mathbf{a}^{\mathbf{m}} \mathbf{b}^{\mathbf{n}}, \sum_{\mathbf{m}, \mathbf{n} \in \mathbf{Z}_{\geq 0}^k} r_{\mathbf{m}, \mathbf{n}} \mathbf{a}^{\mathbf{n}} \mathbf{b}^{\mathbf{m}} \right),$$

where all but finitely many of the $r_{\mathbf{m}, \mathbf{n}} \in R$ are zero. By Corollary 2.12, the congruence E consists of all elements (f, g) for which there exists a transitive chain $(f, r_1), (r_1, r_2), \dots, (r_n, g) \in \mathcal{S}$. Similarly let $\mathcal{T}$ denote the set of all pairs of the form

$$\left(\sum_{\mathbf{m}, \mathbf{n} \in \mathbf{Z}_{\geq 0}^k} s_{\mathbf{m}, \mathbf{n}} \phi(\mathbf{a})^{\mathbf{m}} \phi(\mathbf{b})^{\mathbf{n}}, \sum_{\mathbf{m}, \mathbf{n} \in \mathbf{Z}_{\geq 0}^k} s_{\mathbf{m}, \mathbf{n}} \phi(\mathbf{a})^{\mathbf{n}} \phi(\mathbf{b})^{\mathbf{m}} \right),$$

where all but finitely many of the $s_{\mathbf{m}, \mathbf{n}} \in S$ are zero. (In our notation here, $\phi(\mathbf{a})^{\mathbf{m}} = \phi(a_1)^{m_1} \dots \phi(a_k)^{m_k}$.) Observe that $(\phi \times \phi)(\mathcal{S}) = \mathcal{T}$. Now, by Corollary 2.12 we have

$\langle \mathcal{T} \rangle = \langle (\phi(a_1), \phi(b_1)), \dots, (\phi(a_k), \phi(b_k)) \rangle$ and so it is enough to prove $\langle \mathcal{T} \rangle = \phi_*(E)$. We certainly have $\mathcal{T} \subseteq \phi_*(E)$. Conversely, take any $(s_1, s_2) \in \phi_*(E)$. Then there exists a transitive chain $(s_1, t_1), (t_1, t_2), \dots, (t_n, s_2) \in (\phi \times \phi)(E)$. As each pair in that transitive chain is an element of $(\phi \times \phi)(E)$, each can be written as the image of an element of E , say $(s_1, t_1) = (\phi(f), \phi(h_1)), (t_1, t_2) = (\phi(k_1), \phi(h_2)), \dots, (t_{n-1}, t_n) = (\phi(k_{n-1}), \phi(h_n))$, and $(t_n, s_2) = (\phi(k_n), \phi(g))$. (Note that we cannot assume $(t_1, t_2) = (\phi(h_1), \phi(h_2))$, etc.) Using [Corollary 2.12](#), there exist transitive chains

$$\begin{aligned} (f, r_{11}), (r_{11}, r_{12}), \dots, (r_{1m_1}, h_1) &\in \mathcal{S} \\ (k_1, r_{21}), (r_{21}, r_{22}), \dots, (r_{2m_2}, h_2) &\in \mathcal{S} \\ &\vdots \\ (k_n, r_{(n+1)1}), (r_{(n+1)1}, r_{(n+1)2}), \dots, (r_{(n+1)m_{n+1}}, g) &\in \mathcal{S}. \end{aligned}$$

Applying $\phi \times \phi$ to these transitive chains gives transitive chains

$$\begin{aligned} (s_1, \phi(r_{11})), (\phi(r_{11}), \phi(r_{12})), \dots, (\phi(r_{1m_1}), t_1) &\in \mathcal{T} \\ (t_1, \phi(r_{21})), (\phi(r_{21}), \phi(r_{22})), \dots, (\phi(r_{2m_2}), t_2) &\in \mathcal{T} \\ &\vdots \\ (t_n, \phi(r_{(n+1)1})), (\phi(r_{(n+1)1}), \phi(r_{(n+1)2})), \dots, (\phi(r_{(n+1)m_{n+1}}), s_2) &\in \mathcal{T}. \end{aligned}$$

Concatenating these chains gives a transitive chain in $\mathcal{T}$ that establishes $(s_1, s_2) \in \langle \mathcal{T} \rangle$. Therefore $\phi_*(E) \subseteq \langle \mathcal{T} \rangle$, and hence equality holds. $\square$

Suppose S is a subsemiring of R . Given a congruence F on S , define its **extension** to R to be the congruence $\langle F \rangle_R$. Similarly, given a congruence E on R , define its **restriction** to S to be $E|_S = E \cap (S \times S)$. Note that $E|_S$ is indeed a congruence on S : Properties (E1)–(E3) for $E|_S$ follow immediately from the corresponding properties for E , while properties (I1) and (I2) follow from the corresponding properties for E together with the fact that $S \times S$ is a subsemiring of $R \times R$.

Lemma 2.16. *Suppose $T \subseteq S \subseteq R$ are inclusions of semirings, $\mathcal{S} \subseteq S \times S$ is any subset, and E is a congruence on R . Then:*

- (i) $\langle \langle \mathcal{S} \rangle_S \rangle_R = \langle \mathcal{S} \rangle_R$;
- (ii) $(E|_S)|_T = E|_T$;
- (iii) $(\langle \mathcal{S} \rangle_R)|_S \supseteq \langle \mathcal{S} \rangle_S$; and
- (iv) $\langle E|_S \rangle_R \subseteq E$.

Proof. For (i), observe that the inclusions $\mathcal{S} \subseteq \langle \mathcal{S} \rangle_S \subseteq \langle \mathcal{S} \rangle_R$ induce the inclusion $\langle \mathcal{S} \rangle_R \subseteq \langle \langle \mathcal{S} \rangle_S \rangle_R \subseteq \langle \mathcal{S} \rangle_R$. For (ii), simply observe that $(S \times S) \cap (T \times T) = T \times T$, hence

$(E|_S)|_T = (E \cap (S \times S)) \cap (T \times T) = E \cap (T \times T) = E|_T$. For (iii), the inclusion $\langle \mathcal{S} \rangle_S \subseteq \langle \mathcal{S} \rangle_R$ induces the inclusion $\langle \mathcal{S} \rangle_S \subseteq (\langle \mathcal{S} \rangle_R)|_S$. For (iv), the inclusion $E|_S \subseteq E$ induces the inclusion $\langle E|_S \rangle_R \subseteq E$. $\square$

Example 2.17. The containments in (iii) and (iv) can be proper. Indeed, if we consider the inclusion $\mathbf{Z} \subset \mathbf{Q}$ and the ideal $\langle 2 \rangle \subset \mathbf{Z}$, then we see that $\langle 2 \rangle_{\mathbf{Q}} = \mathbf{Q}$ and so $(\langle 2 \rangle_{\mathbf{Q}})|_{\mathbf{Z}} = \mathbf{Z} \supsetneq \langle 2 \rangle$. For an example of proper containment in (iv), consider the congruence $E = \langle (x, 0_{\mathbf{T}}) \rangle$ on $\mathbf{T}[\mathbf{x}]$. This congruence consists of all pairs (f, g) of tropical polynomials with the same constant term (cf., Example 2.10). It follows that $E|_{\mathbf{T}} = \Delta_{\mathbf{T}}$, and hence $\langle E|_{\mathbf{T}} \rangle_{\mathbf{T}[\mathbf{x}]} = \Delta_{\mathbf{T}[\mathbf{x}]} \subsetneq E$.

Proposition 2.18. *If E is a congruence on a semiring R , then there is a canonical semiring isomorphism*

$$R[x]/\langle E \rangle_{R[x]} \cong (R/E)[x].$$

Proof. Extend the surjective semiring morphism $\pi_E : R \rightarrow R/E$ to a surjective semiring morphism $\tilde{\pi}_E : R[x] \rightarrow (R/E)[x]$ in the natural way, i.e., by defining $\tilde{\pi}_E(\sum_d r_d x^d) = \sum_d \pi_E(r_d) x^d$. Observe that, by construction,

$$\begin{aligned} \ker \tilde{\pi}_E &= \left\{ \left(\sum_d r_d x^d, \sum_d s_d x^d \right) : \sum_d \pi_E(r_d) x^d = \sum_d \pi_E(s_d) x^d \right\} \\ &= \left\{ \left(\sum_d r_d x^d, \sum_d s_d x^d \right) : \pi_E(r_d) = \pi_E(s_d) \text{ for every } d \right\} \\ &= \left\{ \left(\sum_d r_d x^d, \sum_d s_d x^d \right) : (r_d, s_d) \in E \text{ for every } d \right\}. \end{aligned}$$

In particular, we certainly have $E \subseteq \ker \tilde{\pi}_E$ (via pairs of constant polynomials). Moreover, any congruence on $R[x]$ that contains the set E must contain the above collection of pairs, by virtue of the defining properties of a congruence on $R[x]$. Thus, $\ker \tilde{\pi}_E$ is the smallest congruence on $R[x]$ that contains E , and hence $\ker \tilde{\pi}_E = \langle E \rangle_{R[x]}$. $\square$

2.5. Prime and maximal congruences

We say a proper congruence E on a semiring R is **prime** if whenever $(ac, bc) \in E$ for some $a, b, c \in R$, either $(a, b) \in E$ or $(c, 0) \in E$. We say a semiring R is an **integral domain** if whenever $ac = bc$ for some $a, b, c \in R$, either $a = b$ or $c = 0$.

Remark 2.19. We have adopted here the conventional definition of a prime congruences, e.g., as in [23]. By contrast, Joó–Mincheva have recently proposed an alternative definition made with respect to the twisted product. More precisely, in [34] a congruence E

on an additively idempotent semiring R is called prime if whenever $(a, b) \rtimes (c, d) \in E$ one necessarily has either $(a, b) \in E$ or $(c, d) \in E$. For the purpose of this paper, we will call such congruences “twisted-prime.” In [34] it is shown that twisted-prime congruences have very nice algebraic properties, namely, that they are precisely the congruences that are intersection indecomposable with cancellative quotients. Twisted-prime congruences also appear to be especially well adapted to the notion of radical for congruences. See [34] for further details.

As in the case with rings, a proper congruence E on a semiring R is prime if and only if the quotient semiring R/E is an integral domain. By a standard Zorn’s lemma argument, it can be shown that every proper congruence on a semiring is contained in a maximal congruence. Maximal congruences are closely related to fields.

Proposition 2.20. [23, Prop 7.7] *If a semiring R has no nontrivial proper congruences, then either $R \cong \mathbf{B}$ or R is a field.*

Corollary 2.21. *A proper congruence E on a semiring R is maximal if and only if the quotient semiring R/E is isomorphic to either $\mathbf{B}$ or a field.*

Corollary 2.22. *A proper congruence E on an additively idempotent semiring R is maximal if and only if the quotient semiring R/E is isomorphic to $\mathbf{B}$.*

Proof. We always have the semiring surjection $\pi : R \rightarrow R/E$. It follows that

$$1_{R/E} = \pi(1_R) = \pi(1_R + 1_R) = \pi(1_R) + \pi(1_R) = 1_{R/E} + 1_{R/E}.$$

If R/E is a field, then this implies $1_{R/E} = 0_{R/E}$, a contradiction. $\square$

As both $\mathbf{B}$ and fields are integral domains, it follows that every maximal congruence on a semiring is prime, and hence every proper congruence is contained in a prime congruence.

Example 2.23. By Example 2.9, there are precisely two proper congruences on $\mathbf{T}$: the trivial congruence (with factor semiring $\mathbf{T}$) and the congruence $E_{\mathbf{B}}$ (with factor semiring $\mathbf{B}$). As $\mathbf{T}$ is an integral domain, the trivial congruence is prime. Thus, $\mathbf{T}$ has two prime congruences, exactly one of which is maximal.

Example 2.24. Consider the additively idempotent semiring $(\mathbf{N} \cup \{\infty\}, \min, +)$. By Corollary 2.22 the maximal congruences correspond to semiring surjections $\phi : \mathbf{N} \cup \{\infty\} \rightarrow \mathbf{B}$. Such a semiring surjection must satisfy $\phi(\infty) = 0_{\mathbf{B}}$, $\phi(0) = 1_{\mathbf{B}}$, and $\phi(n) = n\phi(1) = \phi(1)$ for every positive $n \in \mathbf{N}$. (Recall that addition in $\mathbf{B}$ is idempotent.) It follows that ϕ is entirely determined by $\phi(1)$, and each of the two possibilities is equally valid. If $\phi(1) = 0_{\mathbf{B}}$, then $\phi(n) = 0_{\mathbf{B}}$ for every nonzero $n \in \mathbf{N} \cup \{\infty\}$. This map corresponds to the maximal

congruence $\langle(1, \infty)\rangle$. If, on the other hand, $\phi(1) = 1_{\mathbf{B}}$, then $\phi(n) = 1_{\mathbf{B}}$ for every $n \in \mathbf{N}$. This map corresponds to the maximal congruence $\langle(0, 1)\rangle$. One can show these are the only nontrivial prime congruences.

Example 2.25. Consider the additively idempotent semiring $(\mathbf{N} \cup \{-\infty\}, \max, +)$. As in the previous example, maximal congruences correspond to semiring surjections onto $\mathbf{B}$. In this case, however, there is only one semiring surjection $\phi : \mathbf{N} \cup \{-\infty\} \rightarrow \mathbf{B}$. Indeed, we must have $\phi(-\infty) = 0_{\mathbf{B}}$, $\phi(0) = 1_{\mathbf{B}}$, and $\phi(1) = \phi(\max\{0, 1\}) = \phi(0) + \phi(1) = 1_{\mathbf{B}} + \phi(1) = 1_{\mathbf{B}}$. It follows that $\phi(n) = 1_{\mathbf{B}}$ for all $n \in \mathbf{N}$, and hence ϕ is completely determined (and corresponds to the maximal congruence $\langle(0, 1)\rangle$). One can show this is the only nontrivial prime congruence.

3. Tropical functions

3.1. Congruences and congruence varieties

We now restrict our attention to the $\mathbf{T}$ -semialgebra $\mathbf{T}[\mathbf{x}] = \mathbf{T}[x_1, \dots, x_n]$. To any subset $S \subseteq \mathbf{T}^n$ we can associate a congruence

$$\mathbf{E}(S) = \{(f, g) \mid f(\mathbf{a}) = g(\mathbf{a}) \text{ for all } \mathbf{a} \in S\} \subseteq \mathbf{T}[\mathbf{x}] \times \mathbf{T}[\mathbf{x}].$$

This association is clearly inclusion-reversing. Conversely, to any subset $E \subseteq \mathbf{T}[\mathbf{x}] \times \mathbf{T}[\mathbf{x}]$ we can associate a set

$$\mathbf{V}(E) = \{\mathbf{a} \in \mathbf{T}^n \mid f(\mathbf{a}) = g(\mathbf{a}) \text{ for all } (f, g) \in E\} \subseteq \mathbf{T}^n.$$

We call $\mathbf{V}(E)$ the **congruence variety** associated to E . This association is also inclusion-reversing. For any congruence E on $\mathbf{T}[\mathbf{x}]$ one has $E \subseteq \mathbf{E}(\mathbf{V}(E))$ and $\mathbf{V}(\mathbf{E}(\mathbf{V}(E))) = \mathbf{V}(E)$. The association $E \mapsto \mathbf{V}(E)$ also behaves well with respect to twisted products:

Lemma 3.1. *For any $(f, g), (f', g') \in \mathbf{T}[\mathbf{x}] \times \mathbf{T}[\mathbf{x}]$,*

$$\mathbf{V}((f, g) \rtimes (f', g')) = \mathbf{V}(f, g) \cup \mathbf{V}(f', g').$$

Proof. As $(f, g) \rtimes (f', g') \in \langle(f, g)\rangle$, we automatically have $\mathbf{V}(f, g) \subseteq \mathbf{V}((f, g) \rtimes (f', g'))$; similarly, $\mathbf{V}(f', g') \subseteq \mathbf{V}((f, g) \rtimes (f', g'))$. Conversely, suppose $\mathbf{a} \in \mathbf{V}((f, g) \rtimes (f', g'))$. The equality

$$((ff') + (gg'))(\mathbf{a}) = ((fg') + (gf'))(\mathbf{a})$$

is equivalent in classical operations to the equality

$$\max\{f(\mathbf{a}) + f'(\mathbf{a}), g(\mathbf{a}) + g'(\mathbf{a})\} = \max\{f(\mathbf{a}) + g'(\mathbf{a}), g(\mathbf{a}) + f'(\mathbf{a})\}.$$

Relabeling if necessary, we may assume $f(\mathbf{a}) \geq g(\mathbf{a})$ and $f'(\mathbf{a}) \geq g'(\mathbf{a})$. The above equality then reduces to

$$f(\mathbf{a}) + f'(\mathbf{a}) = \max\{f(\mathbf{a}) + g'(\mathbf{a}), g(\mathbf{a}) + f'(\mathbf{a})\}.$$

We claim that either $f(\mathbf{a}) = g(\mathbf{a})$ or $f'(\mathbf{a}) = g'(\mathbf{a})$. Suppose not, so that $f(\mathbf{a}) > g(\mathbf{a})$ and $f'(\mathbf{a}) > g'(\mathbf{a})$. Note that, in particular, we must then have $f(\mathbf{a}), f'(\mathbf{a}) > 0_{\mathbf{T}} = -\infty$. It follows, then, that $f(\mathbf{a}) + f'(\mathbf{a}) > f(\mathbf{a}) + g'(\mathbf{a})$ and $f(\mathbf{a}) + f'(\mathbf{a}) > g(\mathbf{a}) + f'(\mathbf{a})$, and so the equality above cannot be satisfied. Thus, we must have either $f(\mathbf{a}) = g(\mathbf{a})$ or $f'(\mathbf{a}) = g'(\mathbf{a})$, and hence $\mathbf{a} \in \mathbf{V}(f, g) \cup \mathbf{V}(f', g')$, as desired. $\square$

In analogy with the classical situation, a congruence variety can be viewed as the set of $\mathbf{T}$ -points of the appropriate semialgebra:

Proposition 3.2. *For any subset $E \subseteq \mathbf{T}[\mathbf{x}] \times \mathbf{T}[\mathbf{x}]$, the set $\mathbf{V}(E)$ is in (natural) bijection with $\text{Hom}_{\mathbf{T}}(\mathbf{T}[\mathbf{x}]/\langle E \rangle, \mathbf{T})$, the set of $\mathbf{T}$ -semialgebra morphisms from $\mathbf{T}[\mathbf{x}]/\langle E \rangle$ to $\mathbf{T}$.*

Proof. First note that each $\mathbf{T}$ -semialgebra morphism $\phi : \mathbf{T}[\mathbf{x}] \rightarrow \mathbf{T}$ is entirely and uniquely determined by $\phi(x_1), \dots, \phi(x_n) \in \mathbf{T}$, i.e., by a point $\mathbf{a} \in \mathbf{T}^n$. This gives the bijection of $\text{Hom}_{\mathbf{T}}(\mathbf{T}[\mathbf{x}], \mathbf{T})$ with $\mathbf{T}^n$; each point $\mathbf{a} \in \mathbf{T}^n$ corresponds to the $\mathbf{T}$ -semialgebra morphism that is evaluation at $\mathbf{a}$. Observe that such a $\mathbf{T}$ -semialgebra morphism $\text{ev}_{\mathbf{a}} : \mathbf{T}[\mathbf{x}] \rightarrow \mathbf{T}$ factors through the associated quotient map $\pi : \mathbf{T}[\mathbf{x}] \rightarrow \mathbf{T}[\mathbf{x}]/\langle E \rangle$ exactly when $\langle E \rangle \subseteq \ker \text{ev}_{\mathbf{a}}$. Since $\ker \text{ev}_{\mathbf{a}}$ is a congruence, we have that $\langle E \rangle \subseteq \ker \text{ev}_{\mathbf{a}}$ if and only if $E \subseteq \ker \text{ev}_{\mathbf{a}}$, i.e., $f(\mathbf{a}) = g(\mathbf{a})$ for every $(f, g) \in E$. This gives the bijection of $\text{Hom}_{\mathbf{T}}(\mathbf{T}[\mathbf{x}]/\langle E \rangle, \mathbf{T})$ with $\mathbf{V}(E)$. $\square$

One consequence of the above result is that $\mathbf{V}(E) = \mathbf{V}(\langle E \rangle)$. So, despite the fact that the general member of a finitely generated congruence E does not have a simple expression in terms of the generators, the above proposition guarantees that the set $\mathbf{V}(E)$ can still be computed from the generators.

Remark 3.3. To each tropical polynomial $f \in \mathbf{T}[\mathbf{x}]$ one can also associate a subset $\text{Bend}(f) \subseteq \mathbf{T}^n$ called the **bend locus** (or **double-max locus**) of f . If $f(\mathbf{x}) = \sum_{\mathbf{d}} c_{\mathbf{d}} \mathbf{x}^{\mathbf{d}}$, then in classical operations $f(\mathbf{x}) = \max_{\mathbf{d}} \{c_{\mathbf{d}} + \mathbf{x} \cdot \mathbf{d}\}$ and $\text{Bend}(f)$ is the collection of points at which the maximum is achieved at least twice. These are the points at which the graph of f “bends.” For example, consider the tropical polynomial $f(x) = x^2 + 2x + 1_{\mathbf{T}}$, which in classical operations is given by $f(x) = \max\{2x, 2 + x, 0\}$. The graph of f bends at $x = 2$ (where the max is achieved by the first and second terms) and at $x = -2$ (where the max is achieved by the second and third terms). Thus, $\text{Bend}(f) = \{-2, 2\}$.

The bend locus is sometimes also called the **corner locus** or **non-differentiability locus** of f . Bend loci are the types of subsets that are usually studied in tropical geometry. While bend loci and congruence varieties are not identical, the bend locus of any single tropical polynomial is a congruence variety. Indeed, one can verify that

$$\text{Bend}(f) = \bigcup_c \mathbf{V} \left(\sum_{\mathbf{d} \in \mathcal{C}} c_{\mathbf{d}} \mathbf{x}^{\mathbf{d}}, \sum_{\mathbf{e} \in \mathcal{C}^{\circ}} c_{\mathbf{e}} \mathbf{x}^{\mathbf{e}} \right) = \mathbf{V} \left(\times_{\mathcal{C}} \left(\sum_{\mathbf{d} \in \mathcal{C}} c_{\mathbf{d}} \mathbf{x}^{\mathbf{d}}, \sum_{\mathbf{e} \in \mathcal{C}^{\circ}} c_{\mathbf{e}} \mathbf{x}^{\mathbf{e}} \right) \right),$$

where the union runs over all nonempty proper subsets of monomials that occur with nonzero coefficient in f , and $\mathcal{C}^{\circ}$ denotes the complementary subset of monomials. For example, in the case of the tropical polynomial $f(x) = x^2 + 2x + 1_{\mathbf{T}}$ considered above, we have

$$\text{Bend}(f) = \mathbf{V}(x^2 + 2x, 1_{\mathbf{T}}) \cup \mathbf{V}(x^2 + 1_{\mathbf{T}}, 2x) \cup \mathbf{V}(2x + 1_{\mathbf{T}}, x^2) = \{-2\} \cup \{-2, 2\} \cup \{2\}.$$

Similarly, one always has $\mathbf{V}(f, g) \subseteq \text{Bend}(f + g)$, so there is a close relationship between congruence varieties and bend loci. For comparison with the above description of the bend locus as a union, we note that one of the key insights in [19] in the definition of scheme-theoretic tropicalization is the observation that tropical varieties can be written as particular intersections; see [19] for complete details.

Question. Can one characterize those congruences that can be written as bend loci?

As in classical algebraic geometry, it is natural to ask in what sense the associations $E \mapsto \mathbf{V}(E)$ and $V \mapsto \mathbf{E}(V)$ establish a correspondence between subsets of $\mathbf{T}^n$ and congruences on $\mathbf{T}[\mathbf{x}]$. In particular, two questions arise. First, when is the set $\mathbf{V}(E)$ nonempty? And secondly, what is the algebraic structure of $\mathbf{E}(\mathbf{V}(E))$? Answers to these questions represent tropical analogues of the weak and strong forms of Hilbert’s Nullstellensatz, respectively. We lay the foundations for these results by analyzing two simple cases: first, the case when $\mathbf{V}(f, g)$ is empty; and second, the case when $\mathbf{V}(f, g) = \mathbf{T}^n$.

3.2. Tropical polynomials that never agree

This section can be viewed as laying the foundation for the weak form of the Tropical Nullstellensatz for Congruences. First, a simple lemma:

Lemma 3.4. *Suppose $f \in \mathbf{T}[\mathbf{x}]$ and $\mathbf{a}, \mathbf{b} \in \mathbf{T}^n$. If $a_i \leq b_i$ for all i , then $f(\mathbf{a}) \leq f(\mathbf{b})$.*

Proof. Say $f(\mathbf{x}) = \sum_{\mathbf{d}} t_{\mathbf{d}} \mathbf{x}^{\mathbf{d}}$. In classical operations, $f(\mathbf{x}) = \max_{\mathbf{d}} \{t_{\mathbf{d}} + \mathbf{d} \cdot \mathbf{x}\}$. If $\mathbf{a}, \mathbf{b} \in \mathbf{T}^n$ are such that $a_i \leq b_i$ for every i , then $\mathbf{d} \cdot \mathbf{a} \leq \mathbf{d} \cdot \mathbf{b}$ for every $\mathbf{d} \in \mathbf{N}^n$. Thus, $f(\mathbf{a}) \leq f(\mathbf{b})$. $\square$

For reasons that will become clear when we prove the weak form of the Tropical Nullstellensatz for Congruences, we now analyze the slightly more general situation in which a pair of functions (f, g) agrees at most in $\mathbf{T}^n \setminus \mathbf{R}^n$.

Lemma 3.5. *Suppose $f, g \in \mathbf{T}[\mathbf{x}]$ are such that $f(\mathbf{a}) \neq g(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{R}^n$. Then either $f(\mathbf{a}) \leq g(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{T}^n$, or $f(\mathbf{a}) \geq g(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{T}^n$. In both cases, equality occurs only when $f(\mathbf{a}) = 0_{\mathbf{T}} = g(\mathbf{a})$.*

Proof. The result is trivially true if either f or g is the zero polynomial, so we may assume f and g are nonzero. First suppose $n = 1$. Observe that $f|_{\mathbf{R}}$ and $g|_{\mathbf{R}}$ are continuous, piecewise-linear real-valued functions. By hypothesis, these functions are never equal, so by the usual Intermediate Value Theorem we must either have $f(a) < g(a)$ for all $a \in \mathbf{R}$, or $f(a) > g(a)$ for all $a \in \mathbf{R}$. We claim that whichever inequality holds, the same inequality (though no longer strict) must hold for $f(0_{\mathbf{T}})$ and $g(0_{\mathbf{T}})$.

Say $f(x) = \sum_{i \geq 0} c_i x^i$ and $g(x) = \sum_{i \geq 0} d_i x^i$. In classical operations, $f(x) = \max_{i \geq 0} \{c_i + ix\}$ and $g(x) = \max_{i \geq 0} \{d_i + ix\}$. Let m_f (resp. m_g) be the least index i for which $c_i > 0_{\mathbf{T}}$ (resp. $d_i > 0_{\mathbf{T}}$). Then there exists some real number M such that $f(a) = c_{m_f} + m_f a$ and $g(a) = d_{m_g} + m_g a$ for all $a \in [-\infty, M] \subset \mathbf{T}$. If $m_f, m_g > 0$, then $f(0_{\mathbf{T}}) = 0_{\mathbf{T}} = g(0_{\mathbf{T}})$. If $m_f = 0$ and $m_g > 0$, then $f(a) = c_0 > g(a)$ for $a \in \mathbf{R}$ sufficiently negative, and $f(0_{\mathbf{T}}) = c_0 > 0_{\mathbf{T}} = g(0_{\mathbf{T}})$. Similarly, if $m_g = 0$ and $m_f > 0$, then $f(a) < d_0 = g(a)$ for $a \in \mathbf{R}$ sufficiently negative, and $f(0_{\mathbf{T}}) = 0_{\mathbf{T}} < d_0 = g(0_{\mathbf{T}})$. Finally, if both $m_f, m_g = 0$, then $f(a) = c_0 = f(0_{\mathbf{T}})$ and $g(a) = d_0 = g(0_{\mathbf{T}})$ for $a \in \mathbf{R}$ sufficiently negative. Thus, in every case we either have $f(0_{\mathbf{T}}) = 0_{\mathbf{T}} = g(0_{\mathbf{T}})$ or the relationship between $f(0_{\mathbf{T}})$ and $g(0_{\mathbf{T}})$ is the same as the relationship between $f(a)$ and $g(a)$ for $a \in \mathbf{R}$ sufficiently negative. We've therefore proven the statement in the case $n = 1$.

Suppose now $n > 1$. The functions $f|_{\mathbf{R}^n}$ and $g|_{\mathbf{R}^n}$ are continuous, piecewise-linear real-valued functions that are never equal, so by the Intermediate Value Theorem we either have $f(\mathbf{a}) < g(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{R}^n$, or $f(\mathbf{a}) > g(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{R}^n$. Without loss of generality, assume we have the former. Now take any $\mathbf{a} \in \mathbf{T}^n \setminus \mathbf{R}^n$, and again without loss of generality suppose $\mathbf{a}$ is of the form $\mathbf{a} = (a_1, \dots, a_m, 0_{\mathbf{T}}, \dots, 0_{\mathbf{T}})$ with $a_i > 0_{\mathbf{T}}$ for $i = 1, \dots, m$. (Let $m = 0$ denote the case where $\mathbf{a}$ is the tropical origin.) Consider the one-variable polynomials $\tilde{f}(s) := f(a_1, \dots, a_m, s, \dots, s)$ and $\tilde{g}(s) := g(a_1, \dots, a_m, s, \dots, s)$. We have $\tilde{f}, \tilde{g} \in \mathbf{T}[s]$ and $\tilde{f}(b) < \tilde{g}(b)$ for every $b \in \mathbf{R}$, so by the one-variable argument above it follows that $f(\mathbf{a}) = \tilde{f}(0_{\mathbf{T}}) \leq \tilde{g}(0_{\mathbf{T}}) = g(\mathbf{a})$, with equality only when $f(\mathbf{a}) = \tilde{f}(0_{\mathbf{T}}) = 0_{\mathbf{T}} = \tilde{g}(0_{\mathbf{T}}) = g(\mathbf{a})$. $\square$

Corollary 3.6. *Suppose $f, g \in \mathbf{T}[\mathbf{x}]$ are such that $f(\mathbf{a}) \neq g(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{T}^n$. Then either $f(\mathbf{a}) < g(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{T}^n$, or $f(\mathbf{a}) > g(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{T}^n$.*

Lemma 3.7. *Suppose $f, g \in \mathbf{T}[\mathbf{x}]$ are such that $f(\mathbf{a}) < g(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{R}^n$. Then there exists some real number $\epsilon > 0$ such that (in tropical operations) $\epsilon f(\mathbf{a}) \leq g(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{T}^n$, with equality only when $f(\mathbf{a}) = 0_{\mathbf{T}} = g(\mathbf{a})$.*

Proof. The result is trivially true if f is the zero polynomial, so we may assume f is nonzero. For the sake of this proof, it is most clear if we use classical operations, so all operations in this proof should be viewed as classical arithmetic. We first consider the case when $n = 1$. Note that if $f(0_{\mathbf{T}}) = 0_{\mathbf{T}} = -\infty$, then we always have $\epsilon + f(0_{\mathbf{T}}) \leq g(0_{\mathbf{T}})$ regardless of ϵ . On the other hand, suppose $f(0_{\mathbf{T}}) > 0_{\mathbf{T}}$ (and hence also $g(0_{\mathbf{T}}) > 0_{\mathbf{T}}$). As f and g are piecewise-linear (with finitely many pieces), there is a real number $M \in \mathbf{R}$ such that f and g are linear on $[-\infty, M] \subset \mathbf{T}$. Under our current assumption that

$f(0_{\mathbf{T}}), g(0_{\mathbf{T}}) > 0_{\mathbf{T}}$, it follows that f and g must actually be constant on $[-\infty, M]$. In particular, $0_{\mathbf{T}} < f(0_{\mathbf{T}}) = f(M) < g(M) = g(0_{\mathbf{T}})$. Now, as the functions $f|_{\mathbf{R}}$ and $g|_{\mathbf{R}}$ are real-valued and piecewise-linear (with finitely many pieces) with $f(a) < g(a)$ always, it follows that

$$\epsilon := \inf_{a \in \mathbf{R}} (g(a) - f(a)) = \min_{a \in \mathbf{R}} (g(a) - f(a)) > 0.$$

Note that we also have $g(0_{\mathbf{T}}) - f(0_{\mathbf{T}}) = g(M) - f(M) \geq \epsilon$. We thus have $\epsilon + f(a) \leq g(a)$ for all $a \in \mathbf{T}$, with equality only when $f(a) = 0_{\mathbf{T}} = g(a)$. We've thus proven the statement for the case $n = 1$.

Suppose now $n > 1$, and observe that the functions $f|_{\mathbf{R}^n}, g|_{\mathbf{R}^n}$ are real-valued and piecewise-linear (with finitely many pieces), with $f(\mathbf{a}) < g(\mathbf{a})$ always. It follows that

$$\epsilon := \inf_{\mathbf{R}^n} (g(\mathbf{a}) - f(\mathbf{a})) = \min_{\mathbf{R}^n} (g(\mathbf{a}) - f(\mathbf{a})) > 0.$$

It follows that $\epsilon + f(\mathbf{a}) \leq g(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{R}^n$. To handle points on the boundary of $\mathbf{T}^n$, we use the same type of argument as in the proof of [Lemma 3.5](#). Take any $\mathbf{a} \in \mathbf{T}^n \setminus \mathbf{R}^n$, and without loss of generality suppose $\mathbf{a}$ is of the form $\mathbf{a} = (a_1, \dots, a_m, 0_{\mathbf{T}}, \dots, 0_{\mathbf{T}})$ with $a_i > 0_{\mathbf{T}}$ for $i = 1, \dots, m$. (Let $m = 0$ denote the case where $\mathbf{a}$ is the tropical origin.) Consider the one-variable polynomials $\tilde{f}(s) := f(a_1, \dots, a_m, s, \dots, s)$ and $\tilde{g}(s) := g(a_1, \dots, a_m, s, \dots, s)$. We have $\tilde{g}(b) - \tilde{f}(b) \geq \epsilon$ for every $b \in \mathbf{R}$, so by the one-dimensional argument above we have $\epsilon + \tilde{f}(b) \leq \tilde{g}(b)$ for all $b \in \mathbf{T}$, with equality only when $\tilde{f}(b) = 0_{\mathbf{T}} = \tilde{g}(b)$. It follows that $\epsilon + f(\mathbf{a}) = \epsilon + \tilde{f}(0_{\mathbf{T}}) \leq \tilde{g}(0_{\mathbf{T}}) = g(\mathbf{a})$, with equality only when $f(\mathbf{a}) = 0_{\mathbf{T}} = g(\mathbf{a})$, as desired. $\square$

Corollary 3.8. *Suppose $f, g \in \mathbf{T}[\mathbf{x}]$ are such that $f(\mathbf{a}) < g(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{T}^n$. Then there exists some real number $\epsilon > 0$ such that (in tropical operations) $\epsilon f(\mathbf{a}) < g(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{T}^n$.*

Corollary 3.9. *Suppose $f, g \in \mathbf{T}[\mathbf{x}]$ satisfy $\mathbf{V}(f, g) \subseteq \mathbf{T}^n \setminus \mathbf{R}^n$, and without loss of generality (in view of [Lemma 3.5](#)) assume $f(\mathbf{a}) \leq g(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{T}^n$. Then $(g, \epsilon g) \in \langle (f, g), \mathbf{E}(\mathbf{T}^n) \rangle$ for some real number $\epsilon > 0$.*

Proof. By [Lemmas 3.5 and 3.7](#), there exists $\epsilon > 0$ such that $\epsilon f(\mathbf{a}) \leq g(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{T}^n$. As tropical addition corresponds to real maximum, it follows that $g + \epsilon f$ and g define the same polynomial functions, i.e., $(g + \epsilon f, g) \in \mathbf{E}(\mathbf{T}^n)$. As $(\epsilon f, \epsilon g) \in \langle (f, g) \rangle$, we also have $(g + \epsilon f, \epsilon g) = (g, g) + (\epsilon f, \epsilon g) \in \langle (f, g) \rangle$. By transitivity, then, we have $(g, \epsilon g) \in \langle (f, g), \mathbf{E}(\mathbf{T}^n) \rangle$. $\square$

Corollary 3.10. *The congruence $\mathbf{E}(\mathbf{T}^n)$ is prime.*

Proof. Suppose $(hf, hg) \in \mathbf{E}(\mathbf{T}^n)$ for some $h, f, g \in \mathbf{T}[\mathbf{x}]$. If h is the zero polynomial, then it is the zero function on $\mathbf{T}^n$, and so $(h, 0_{\mathbf{T}}) \in \mathbf{E}(\mathbf{T}^n)$. Suppose now h is not the

zero polynomial. Then we have $h(\mathbf{a}) > 0_{\mathbf{T}}$ for all $\mathbf{a} \in \mathbf{R}^n$, and hence $h(\mathbf{a})$ has a (tropical) multiplicative inverse for every $\mathbf{a} \in \mathbf{R}^n$. As $h(\mathbf{a})f(\mathbf{a}) = h(\mathbf{a})g(\mathbf{a})$ for all $\mathbf{a}$, this implies $f(\mathbf{a}) = g(\mathbf{a})$ for all $\mathbf{a} \in \mathbf{R}^n$. We claim it follows that $f(\mathbf{a}) = g(\mathbf{a})$ for all $\mathbf{a} \in \mathbf{T}^n$. By the same argument used in both Lemma 3.5 and Lemma 3.7, one can reduce to the one-variable case, where the result is clear. $\square$

We call the $\mathbf{T}$ -semialgebra $\mathbf{T}[\mathbf{x}]/\mathbf{E}(\mathbf{T}^n)$ the **semiring of tropical polynomial functions**, denoted $\text{Poly}[\mathbf{T}^n]$. By the above corollary, it is an integral domain.

3.3. Tropical polynomials that always agree

We next analyze the situation in which two tropical polynomials agree everywhere, and hence define the same function on $\mathbf{T}^n$. This section can be viewed as laying the groundwork for the strong form of the Tropical Nullstellensatz for Congruences. Given a tropical polynomial $f \in \mathbf{T}[\mathbf{x}]$, write $[f]$ for the equivalence class of tropical polynomials that define the same function on $\mathbf{T}^n$, i.e., let $[f]$ denote the image of f in $\text{Poly}[\mathbf{T}^n] = \mathbf{T}[\mathbf{x}]/\mathbf{E}(\mathbf{T}^n)$. We will identify two “extremal” representatives in $[f]$, which we refer to as the *desaturated* and *saturated* representatives of $[f]$.

3.3.1. Desaturated polynomials

The “minimal” representative of $[f]$ is the unique tropical polynomial f_{dsat} whose coefficients are each minimal among all tropical polynomials representing the function $[f]$. (As tropical addition corresponds to real max, this is well defined and can be computed independently for each coefficient.) We call f_{dsat} the **desaturated representative** of $[f]$. This polynomial can be explicitly computed as follows. First choose any representative polynomial for $[f]$, say $f(\mathbf{x}) = \sum_{\mathbf{d}} a_{\mathbf{d}} \mathbf{x}^{\mathbf{d}}$; in classical operations, we then have $[f(\mathbf{x})] = \max_{\mathbf{d}} \{a_{\mathbf{d}} + \mathbf{d} \cdot \mathbf{x}\}$. The graph of $[f]$ is then given by the upper convex hull of the union of the hyperplanes $\{z = a_{\mathbf{d}} + \mathbf{d} \cdot \mathbf{x}\} \subset \mathbf{T}^{n+1}$. (Here the equations for the hyperplanes are given in classical operations.) The projection of this graph onto $\mathbf{T}^n$ gives a finite decomposition of $\mathbf{T}^n$ into closed n -dimensional polyhedral regions, in the interior of each of which $[f]$ is given by a single affine linear function (one of the monomial terms in the tropical polynomial). The boundaries of these regions are where two (or more) of the monomial terms restrict to the same affine linear function. In particular, the union of the codimension one faces is precisely the double-max locus of f . We say the monomials corresponding to the top-dimensional polyhedra in this decomposition **contribute** to the function $[f]$. For each index $\mathbf{d} \in \mathbf{Z}_{\geq 0}^n$, we set $\underline{a}_{\mathbf{d}} = a_{\mathbf{d}}$ if the $\mathbf{d}$ th monomial contributes to $[f]$, and $\underline{a}_{\mathbf{d}} = 0_{\mathbf{T}}$ otherwise. Then

$$f_{\text{dsat}}(\mathbf{x}) = \sum_{\mathbf{d}} \underline{a}_{\mathbf{d}} \mathbf{x}^{\mathbf{d}}.$$

If $g(\mathbf{x}) = \sum_{\mathbf{d}} b_{\mathbf{d}} \mathbf{x}^{\mathbf{d}}$ is any other tropical polynomial with $[g] = [f]$, then $b_{\mathbf{d}} \geq \underline{a}_{\mathbf{d}}$ for every index $\mathbf{d}$, with equality for those indices that contribute to $[f]$. The desaturation

of f is the unique tropical polynomial with this property. We say f is desaturated if $f = f_{\text{dsat}}$.

Example 3.11. Consider the tropical polynomial $f(x, y) = x^3 + xy + 2y^2 + 1$. One can check $f_{\text{dsat}}(x, y) = x^3 + 2y^2 + 1$.

Lemma 3.12. *If $f(\mathbf{x}) = \sum_{\mathbf{d}} \mathbf{a}_{\mathbf{d}} \mathbf{x}^{\mathbf{d}}$ is desaturated, then $(f^m)_{\text{dsat}}(\mathbf{x}) = \sum_{\mathbf{d}} \mathbf{a}_{\mathbf{d}}^m \mathbf{x}^{m\mathbf{d}}$.*

Proof. In classical operations, we have $[f^m] = m[f]$. The function $[f^m]$ therefore induces the same polyhedral decomposition of $\mathbf{T}^n$ as $[f]$, with the affine linear function on each cell equaling m times the affine linear function for $[f]$ on that cell. Thus, if $[f]$ is given (in classical operations) by $a_{\mathbf{d}} + \mathbf{d} \cdot \mathbf{x}$ on a given cell, then $[f^m]$ is given (in classical operations) by $ma_{\mathbf{d}} + m\mathbf{d} \cdot \mathbf{x}$ on that cell. The result follows. $\square$

3.3.2. Saturated polynomials

The “maximal” representative of $[f]$ is the unique tropical polynomial f_{sat} whose coefficients are maximal among all tropical polynomials representing $[f]$. We call f_{sat} the **saturated representative** of $[f]$, and denote its coefficients by $\bar{a}_{\mathbf{d}}$. Note that if $\bar{a}_{\mathbf{d}} > 0_{\mathbf{T}}$, then there always exists some point $\mathbf{t} \in \mathbf{R}^n$ with $f(\mathbf{t}) = \bar{a}_{\mathbf{d}} \mathbf{t}^{\mathbf{d}}$. If $g(\mathbf{x}) = \sum_{\mathbf{d}} b_{\mathbf{d}} \mathbf{x}^{\mathbf{d}}$ is any other tropical polynomial with $[g] = [f]$, then $b_{\mathbf{d}} \leq \bar{a}_{\mathbf{d}}$ for every $\mathbf{d}$. The saturation of f is the unique tropical polynomial with this property. We say f is saturated if $f = f_{\text{sat}}$.

Before explaining how to explicitly calculate f_{sat} , we first introduce some notation. Let $\mathcal{C} = \{\mathbf{d}_1, \dots, \mathbf{d}_k\}$ denote the set of indices contributing to $[f]$. For each $\mathbf{d} \in \mathcal{C}$, let $Z_{\mathbf{d}} \subseteq \mathbf{T}^n$ denote the closed n -dimensional polyhedral region on which $[f]$ is given by $a_{\mathbf{d}} \mathbf{x}^{\mathbf{d}}$. For each (real) convex combination $\sum_{i=1}^k \lambda_i \mathbf{d}_i$, let Z_{λ} denote the intersection of those polyhedral regions $Z_{\mathbf{d}_i}$ for which $\lambda_i > 0$, and define (in tropical operations)

$$\tilde{a}_{\lambda} = \prod_{i=1}^k a_{\mathbf{d}_i}^{\lambda_i}.$$

Lastly, considering $\mathcal{C}$ as a collection of lattice points in $\mathbf{R}^n$, let Δ denote the convex hull of $\mathcal{C}$. We can now give a recipe for computing f_{sat} .

Proposition 3.13. *Let $f \in \mathbf{T}[\mathbf{x}]$ be a tropical polynomial, and let $f_{\text{sat}}(\mathbf{x}) = \sum_{\mathbf{d}} \bar{a}_{\mathbf{d}} \mathbf{x}^{\mathbf{d}}$. For each index $\mathbf{d}$, the following hold:*

- i) if $\mathbf{d} \notin \Delta$, then $\bar{a}_{\mathbf{d}} = 0_{\mathbf{T}}$; and
- ii) if $\mathbf{d} \in \Delta$, then $\bar{a}_{\mathbf{d}} \geq \tilde{a}_{\lambda}$ for any (real) convex combination $\mathbf{d} = \sum_{i=1}^k \lambda_i \mathbf{d}_i$, with equality if $Z_{\lambda} \neq \emptyset$. Moreover, there exists at least one such convex combination with $Z_{\lambda} \neq \emptyset$.

Proof. We will use classical operations for the entirety of this proof. Suppose $\mathbf{d}$ is any index such that $\bar{a}_{\mathbf{d}} > 0_{\mathbf{T}}$. Then there exists some $\mathbf{t}_0 \in \mathbf{R}^n$ with $f(\mathbf{t}_0) = \bar{a}_{\mathbf{d}} + \mathbf{d} \cdot \mathbf{t}_0$.

Let $\mathcal{C}' \subseteq \mathcal{C}$ be the collection of those indices $\mathbf{d}_i$ for which $\mathbf{t}_0 \in Z_{\mathbf{d}_i}$, and let $\Delta' \subseteq \Delta$ be the convex hull of $\mathcal{C}'$. We first prove that every hyperplane in $\mathbf{R}^n$ containing $\mathbf{d}$ intersects Δ' . Let H be any hyperplane containing $\mathbf{d}$, with normal vector $\mathbf{v} \in \mathbf{R}^n$. Since the polyhedra $\{Z_{\mathbf{d}'} \mid \mathbf{d}' \in \mathcal{C}'\}$ give a local decomposition of $\mathbf{R}^n$ near $\mathbf{t}_0$, for sufficiently small $\mu > 0$ the vector $\mathbf{t}_0 + \mu\mathbf{v}$ is contained in (at least) one of the polyhedra $Z_{\mathbf{d}'}$, i.e., the direction $\mathbf{v}$ points into the region $Z_{\mathbf{d}'}$. By the definition of $\bar{a}_{\mathbf{d}}$, we must then have $\bar{a}_{\mathbf{d}} + \mathbf{d} \cdot (\mathbf{t}_0 + \mu\mathbf{v}) \leq f(\mathbf{t}_0 + \mu\mathbf{v}) = \underline{a}_{\mathbf{d}'} + \mathbf{d}' \cdot (\mathbf{t}_0 + \mu\mathbf{v})$ for all sufficiently small $\mu > 0$, and consequently for all $\mu > 0$. In other words, the hyperplane $z = \bar{a}_{\mathbf{d}} + \mathbf{d} \cdot \mathbf{x}$ must lie beneath the hyperplane $z = \underline{a}_{\mathbf{d}'} + \mathbf{d}' \cdot \mathbf{x}$ in the direction $\mathbf{v}$ from the point $\mathbf{t}_0$. Now consider the sequence of points $\mathbf{x}_n = \mathbf{t}_0 + n\mathbf{v}$ for $n \in \mathbf{N}$. By the above discussion, we must have $\bar{a}_{\mathbf{d}} + \mathbf{d} \cdot \mathbf{x}_n \leq \underline{a}_{\mathbf{d}'} + \mathbf{d}' \cdot \mathbf{x}_n$ for every $n \in \mathbf{N}$, and hence

$$\bar{a}_{\mathbf{d}} + \mathbf{d} \cdot \mathbf{t}_0 + n(\mathbf{d} \cdot \mathbf{v}) \leq \underline{a}_{\mathbf{d}'} + \mathbf{d}' \cdot \mathbf{t}_0 + n(\mathbf{d}' \cdot \mathbf{v})$$

for all $n \in \mathbf{N}$. The coefficient $\bar{a}_{\mathbf{d}}$ is real (by hypothesis), as is the coefficient $\underline{a}_{\mathbf{d}'}$ (since $\mathbf{d}' \in \mathcal{C}$), and the point $\mathbf{t}_0 \in \mathbf{R}^n$, so the above inequality involves only real (finite) numbers. Since it holds for all n , we must therefore have $\mathbf{d} \cdot \mathbf{v} \leq \mathbf{d}' \cdot \mathbf{v}$. Repeating this argument in the direction $-\mathbf{v}$, we can similarly deduce there must exist some index $\mathbf{d}'' \in \mathcal{C}'$ with $\mathbf{d}'' \cdot \mathbf{v} \leq \mathbf{d} \cdot \mathbf{v}$. It follows that there exists $\lambda \in [0, 1]$ such that $(\lambda\mathbf{d}' + (1 - \lambda)\mathbf{d}'') \cdot \mathbf{v} = \mathbf{d} \cdot \mathbf{v}$, i.e., $\lambda\mathbf{d}' + (1 - \lambda)\mathbf{d}'' \in H$. Notice that $\mathbf{d}', \mathbf{d}'' \in \mathcal{C}' \subseteq \Delta'$, so by convexity we also have $\lambda\mathbf{d}' + (1 - \lambda)\mathbf{d}'' \in \Delta'$. We've thus proven that every hyperplane containing $\mathbf{d}$ intersects Δ' . As Δ' is convex, this implies $\mathbf{d} \in \Delta' \subseteq \Delta$. This proves (i). Note that it also proves that if $\bar{a}_{\mathbf{d}} > 0_{\mathbf{T}}$, then $\mathbf{d}$ can be written as a convex combination $\mathbf{d} = \sum_i \lambda_i \mathbf{d}_i$ with $Z_{\lambda} \neq \emptyset$.

Conversely, suppose $\mathbf{d} \in \Delta$ and write $\mathbf{d}$ as convex combination $\mathbf{d} = \sum_i \lambda_i \mathbf{d}_i$. We claim that we must then have $\bar{a}_{\mathbf{d}} \geq \tilde{a}_{\lambda}$. Indeed, an inequality $\bar{a}_{\mathbf{d}} < \tilde{a}_{\lambda}$ would imply the existence of a point $\mathbf{t} \in \mathbf{T}^n$ with $\tilde{a}_{\lambda} + \mathbf{d} \cdot \mathbf{t} > f(\mathbf{t})$. But then we would have

$$f(\mathbf{t}) < \tilde{a}_{\lambda} + \mathbf{d} \cdot \mathbf{t} = \sum_{i=1}^k \lambda_i (\underline{a}_{\mathbf{d}_i} + \mathbf{d}_i \cdot \mathbf{t}) \leq \max_{1 \leq i \leq k} (\underline{a}_{\mathbf{d}_i} + \mathbf{d}_i \cdot \mathbf{t}) = f(\mathbf{t}),$$

a contradiction. Thus, we must have $\bar{a}_{\mathbf{d}} \geq \tilde{a}_{\lambda} > 0_{\mathbf{T}}$. By our argument above, it then follows that $\mathbf{d}$ can be written as a convex combination $\mathbf{d} = \sum_i \lambda_i \mathbf{d}_i$ with $Z_{\lambda} \neq \emptyset$. Note that since Z_{λ} is nonempty, it must contain a point $\mathbf{t}_0 \in \mathbf{R}^n$ (by considering the construction of the polyhedral decomposition of $\mathbf{T}^n$). Then observe that

$$\tilde{a}_{\lambda} + \mathbf{d} \cdot \mathbf{t}_0 = \sum_{i=1}^k \lambda_i (\underline{a}_{\mathbf{d}_i} + \mathbf{d}_i \cdot \mathbf{t}_0) = \sum_{i=1}^k \lambda_i f(\mathbf{t}_0) = f(\mathbf{t}_0).$$

As $\mathbf{t}_0 \in \mathbf{R}^n$, this is an equality of real numbers, and hence implies $\bar{a}_{\mathbf{d}} \leq \tilde{a}_{\lambda}$. Thus, equality holds. $\square$

Remark 3.14. Note that for $\mathbf{d} \in \mathcal{C}$ the above proposition guarantees $\bar{a}_{\mathbf{d}} = \underline{a}_{\mathbf{d}}$, as expected.

Example 3.15. Consider the desaturated tropical polynomial $f(x) = x^2 + 4$. Using the above proposition, one computes $f_{\text{sat}}(x) = x^2 + 2x + 4$.

Example 3.16. Fix some $s > t > 1_{\mathbf{T}}$, and consider the desaturated tropical polynomial $f(x, y) = x^3 + sy^2 + t$. Proposition 3.13 gives a straightforward recipe for computing the coefficients of f_{sat} . For example, to compute the coefficient of xy in f_{sat} , we first observe that $(1, 1) = \frac{1}{6}(0, 0) + \frac{1}{3}(3, 0) + \frac{1}{2}(0, 2)$, and that $Z_{(0,0)} \cap Z_{(3,0)} \cap Z_{(0,2)} \neq \emptyset$. It follows that the coefficient of xy in f_{sat} is (in tropical operations) $t^{1/6} \cdot 1_{\mathbf{T}}^{1/3} \cdot s^{1/2}$. The other coefficients are found similarly. In the end, one computes $f_{\text{sat}}(x, y) = x^3 + t^{1/3}x^2 + t^{1/6}s^{1/2}xy + sy^2 + t^{2/3}x + t^{1/2}s^{1/2}y + t$.

Proposition 3.17. Suppose $f \in \mathbf{T}[\mathbf{x}]$ is nonzero, and $k = |\mathcal{C}|$ is the number of monomials that contribute to $[f]$. Then $(f^m)_{\text{sat}}f^m$ is saturated for every $m \geq k$.

Proof. First observe that every coefficient in $(f^m_{\text{dsat}})_{\text{sat}}f^m_{\text{dsat}}$ is less than or equal to the corresponding coefficient in $(f^m)_{\text{sat}}f^m$, and hence the latter polynomial is automatically saturated if the former is. Without loss of generality, we may therefore assume f is desaturated. Next, observe that by Lemma 3.12 we have

$$((f^m)_{\text{sat}}f^m)_{\text{dsat}} = (f^{2m})_{\text{dsat}} = \sum_{\mathbf{d}} \underline{a}_{\mathbf{d}}^{2m} \mathbf{x}^{2m\mathbf{d}},$$

and hence the convex hull of the Newton polytope for $((f^m)_{\text{sat}}f^m)_{\text{dsat}}$ is precisely $2m\Delta$, where Δ is the convex hull of the Newton polytope for f_{dsat} .

We now compute the coefficients in $(f^m)_{\text{sat}}f^m$ and prove they agree with those of its saturation. By Proposition 3.13 and the above observation, we may restrict our attention to indices $\mathbf{e} \in 2m\Delta$. Fix any index $\mathbf{e} \in 2m\Delta$. By Proposition 3.13, the coefficient of $\mathbf{x}^{\mathbf{e}}$ in $((f^m)_{\text{sat}}f^m)_{\text{sat}}$ equals $\prod_i \underline{a}_{\mathbf{d}_i}^{2m\lambda_i}$ for any (real) convex combination $\mathbf{e} = \sum_i \lambda_i(2m\mathbf{d}_i)$ with $Z_{\lambda} \neq \emptyset$. Fix some such convex combination. To compute the coefficient of $\mathbf{x}^{\mathbf{e}}$ in $(f^m)_{\text{sat}}f^m$, first observe that (again by Proposition 3.13)

$$(f^m)_{\text{sat}} = \sum_{\mathbf{d} \in m\Delta} b_{\mathbf{d}} \mathbf{x}^{\mathbf{d}},$$

where $b_{\mathbf{d}} = \prod_i \underline{a}_{\mathbf{d}_i}^{m\mu_i}$ for any (real) convex combination $\mathbf{d} = \sum_i \mu_i(m\mathbf{d}_i)$ with $Z_{\mu} \neq \emptyset$. Next observe that, by direct calculation,

$$f^m = \sum_{\mathbf{d}' \in m\Delta} c_{\mathbf{d}'} \mathbf{x}^{\mathbf{d}'},$$

where $c_{\mathbf{d}'} = \sum_{\mathbf{n} \in \mathcal{S}_{\mathbf{d}'}} \underline{a}_{\mathbf{d}_i}^{n_i}$ (in tropical operations) and where $\mathcal{S}_{\mathbf{d}'} = \{\mathbf{n} \in \mathbf{N}_0^k \mid \mathbf{d}' = \sum_i n_i \mathbf{d}_i, m = \sum_i n_i\}$. (Here $c_{\mathbf{d}'} = 0_{\mathbf{T}}$ if $\mathcal{S}_{\mathbf{d}'}$ is empty.) It follows that the coefficient of $\mathbf{x}^{\mathbf{e}}$ in $(f^m)_{\text{sat}}f^m$ is

$$d_{\mathbf{e}} = \sum_{(\mu, \mathbf{n}) \in \mathcal{T}_{\mathbf{e}}} \prod_{i=1}^k a_{\mathbf{d}_i}^{(m\mu_i + n_i)},$$

where $\mathcal{T}_{\mathbf{e}}$ is the collection of all pairs $(\mu, \mathbf{n}) \in [0, 1]^k \times \mathbf{N}_0^k$ such that $\sum_i \mu_i = 1$, $Z_{\mu} \neq \emptyset$, $\sum_i n_i = m$, and $\mathbf{e} = \sum_i (m\mu_i + n_i) \mathbf{d}_i$, and where $d_{\mathbf{e}} = 0_{\mathbf{T}}$ if $\mathcal{T}_{\mathbf{e}}$ is empty.

We claim $\mathcal{T}_{\mathbf{e}}$ is not empty (and here we will use the hypothesis $m \geq k$). For each $i = 1, \dots, k$, define ${}_0\tau_i = 2m\lambda_i$ and ${}_0n_i = 0$. Observe then that $\sum_{i=1}^k {}_0\tau_i = 2m \geq k$, so there must exist at least one index i_1 with ${}_0\tau_{i_1} \geq 1$. Now define

$${}_1n_i = \begin{cases} {}_0n_i, & \text{if } i \neq i_1 \\ {}_0n_i + 1, & \text{if } i = i_1 \end{cases} \quad \text{and} \quad {}_1\tau_i = \begin{cases} {}_0\tau_i, & \text{if } i \neq i_1 \\ {}_0\tau_i - 1, & \text{if } i = i_1 \end{cases}.$$

Observe that $2m\lambda_i = {}_1\tau_i + {}_1n_i$ for every i , $\sum_i {}_1n_i = 1$, and that $\sum_{i=1}^k {}_1\tau_i = 2m - 1 \geq k$. The last inequality guarantees that there again exists at least one index i_2 with ${}_1\tau_{i_2} \geq 1$. We thus inductively continue, at each stage $0 \leq l \leq m$ producing nonnegative real numbers ${}_l\tau_i$ and nonnegative integers ${}_ln_i$ with $2m\lambda_i = {}_l\tau_i + {}_ln_i$, $\sum_{i=1}^k {}_ln_i = l$, and $\sum_{i=1}^k {}_l\tau_i = 2m - l \geq k$. In particular, at stage m we produce nonnegative real numbers ${}_m\tau_i$ and nonnegative integers ${}_mn_i$ with $2m\lambda_i = {}_m\tau_i + {}_mn_i$, $\sum_{i=1}^k {}_mn_i = m$ and $\sum_{i=1}^k {}_m\tau_i = m$. In other words, if we let $\mu_i = {}_m\tau_i/m$ and $n_i = {}_mn_i$, then $\mathbf{e} = \sum_i (m\mu_i + n_i) \mathbf{d}_i$. Moreover, by construction we have $\mu_i > 0$ only if $\lambda_i > 0$, and hence $Z_{\lambda} \subseteq Z_{\mu}$, i.e., Z_{μ} must be nonempty. We've thus constructed a pair $(\mu, \mathbf{n}) \in \mathcal{T}_{\mathbf{e}}$, and hence proven $\mathcal{T}_{\mathbf{e}}$ is nonempty.

In fact, our construction has proven slightly more. We have actually proven the existence of a pair $(\mu, \mathbf{n}) \in \mathcal{T}_{\mathbf{e}}$ with $2m\lambda = m\mu + \mathbf{n}$. For this pair, we have $\prod_{i=1}^k a_{\mathbf{d}_i}^{m\mu_i + n_i} = \prod_{i=1}^k a_{\mathbf{d}_i}^{2m\lambda_i}$, and hence $d_{\mathbf{e}} \geq \prod_{i=1}^k a_{\mathbf{d}_i}^{2m\lambda_i}$. Since the right-hand side represents the maximum possible value for the coefficient of $\mathbf{x}^{\mathbf{e}}$ in any representative of $[(f^m)_{\text{sat}} f^m]$, we must actually have equality. Thus, $(f^m)_{\text{sat}} f^m$ is saturated. $\square$

Corollary 3.18. *Suppose $f \in \mathbf{T}[\mathbf{x}]$ is nonzero and $k = |\mathcal{C}|$ is the number of monomials that contribute to $[f]$. Then for every $m \geq k$ and every choice of tropical polynomials $f_0, \dots, f_m \in [f]$, the polynomial $(f_0^m)_{\text{sat}} f_1 \cdots f_m$ is saturated.*

Proof. Fix some $f \in \mathbf{T}[\mathbf{x}]$, $m \geq k$, and $f_0, \dots, f_m \in [f]$. First observe that, by applying Proposition 3.17 to the polynomial f_{dsat} , the polynomial $(f_{\text{dsat}}^m)_{\text{sat}} f_{\text{dsat}}^m$ is saturated. Since $(f_0^m)_{\text{sat}} = (f_{\text{dsat}}^m)_{\text{sat}}$, we thus also have that $(f_0^m)_{\text{sat}} f_{\text{dsat}}^m$ is saturated. Now observe that every coefficient in f_{dsat}^m is less than or equal to the corresponding coefficient in $f_1 \cdots f_m$, and hence every coefficient in $(f_0^m)_{\text{sat}} f_{\text{dsat}}^m$ is less than or equal to the corresponding coefficient in $(f_0^m)_{\text{sat}} f_1 \cdots f_m$. Since they both define the same function on $\mathbf{T}^n$, and since $(f_0^m)_{\text{sat}} f_{\text{dsat}}^m$ is saturated (i.e., all coefficients are maximal), it follows that $(f_0^m)_{\text{sat}} f_1 \cdots f_m$ is also saturated. $\square$

Corollary 3.19. Suppose $(f, g) \in \mathbf{E}(\mathbf{T}^n)$ and $k = |\mathcal{C}|$ is the number of monomials that contribute to $[f] = [g]$. Then $(f^m)_{\text{sat}} f^m = (f^m)_{\text{sat}} g^m$, for every $m \geq \max\{1, k\}$.

Proof. Fix some $(f, g) \in \mathbf{E}(\mathbf{T}^n)$ and $m \geq \max\{1, k\}$. Then $g \in [f]$, so by the previous corollary it follows that $(f^m)_{\text{sat}} f^m$ and $(f^m)_{\text{sat}} g^m$ are both saturated. As they both define the same function on $\mathbf{T}^n$, it follows (by the uniqueness of saturated representatives) that they must be equal. $\square$

Example 3.20. Consider the tropical polynomials $f(x) = x^2 + 4$ and $g(x) = x^2 + 1x + 4$. By Example 3.15, we see that $f_{\text{sat}}(x) = g_{\text{sat}}(x) = x^2 + 2x + 4$, and hence $(f, g) \in \mathbf{E}(\mathbf{T}^n)$. As f is desaturated, the number of monomials that contribute to $[f]$ is two, and hence by the above corollary we must have $f_{\text{sat}}^2 f^2 = f_{\text{sat}}^2 g^2$. This is easily verified by direct computation. In fact, one can verify in this case that $f_{\text{sat}} f = f_{\text{sat}} g$.

We can now describe $\mathbf{E}(\mathbf{V}(E))$ in the special case of the trivial congruence $E = \Delta_{\mathbf{T}[x]}$:

Theorem 1. The congruence $\mathbf{E}(\mathbf{T}^n)$ is the intersection of all prime congruences on $\mathbf{T}[\mathbf{x}]$.

Proof. By Corollary 3.10, the congruence $\mathbf{E}(\mathbf{T}^n)$ is prime, and hence automatically contains the intersection of all prime congruences. Conversely, suppose P is any prime congruence on $\mathbf{T}[\mathbf{x}]$, suppose $(f, g) \in \mathbf{E}(\mathbf{T}^n)$, and suppose, by way of contradiction, $(f, g) \notin P$. By Corollary 3.19, we have $((f^m)_{\text{sat}} f^m, (f^m)_{\text{sat}} g^m) \in \Delta_{\mathbf{T}[x]} \subset P$ for any fixed $m \geq \max\{1, k\}$, where k is the number of monomials that contribute to $[f] = [g]$. By primeness (together with the supposition $(f, g) \notin P$) this implies $((f^m)_{\text{sat}}, 0_{\mathbf{T}}) \in P$. We then have $((f^m)_{\text{sat}}, f^m) = (f^m, f^m) + ((f^m)_{\text{sat}}, 0_{\mathbf{T}}) \in P$, hence by transitivity $(f^m, 0_{\mathbf{T}}) \in P$, and hence by primeness $(f, 0_{\mathbf{T}}) \in P$. Similarly, $((f^m)_{\text{sat}}, g^m) = (g^m, g^m) + ((f^m)_{\text{sat}}, 0_{\mathbf{T}}) \in P$, hence $(g^m, 0_{\mathbf{T}}) \in P$, hence $(g, 0_{\mathbf{T}}) \in P$. By transitivity, we then have $(f, g) \in P$, a contradiction. $\square$

Remark 3.21. The classical Nullstellensatz of Hilbert states that if k is an algebraically closed field, then for every ideal I in $k[\mathbf{x}]$ one has $\mathbf{I}(\mathbf{V}(I)) = \sqrt{I}$, where $\sqrt{I}$ is the **radical** of I . By definition, $\sqrt{I}$ is the ideal in $k[\mathbf{x}]$ consisting of those elements f such that $f^n \in I$ for some $n \geq 1$. One of the basic algebraic properties of the radical is that $\sqrt{I}$ is also equal to the intersection of all prime ideals in $k[\mathbf{x}]$ that contain I . In the special case of the trivial ideal in $k[\mathbf{x}]$, Hilbert's Nullstellensatz is thus equivalent to the statement that $\mathbf{I}(k^n)$ is the intersection of all prime ideals in $k[\mathbf{x}]$. Theorem 1 can be seen as a direct tropical analogue of this statement.

Remark 3.22. The above result was recently proved in [34, Prop. 5.4] by completely different methods.

Remark 3.23. The difference between tropical polynomials and tropical functions has recently been explored in [45]. In that paper, the notion of integral closure for rings is

extended to semirings, and it is shown that in the case of a free semiring the notion of integrally closed for an element coincides with the notion of saturated defined here.

4. The weak tropical Nullstellensatz for congruences

We now aim to understand the general situation in which $\mathbf{V}(E)$ is empty. We begin by examining the case in which E is principally generated.

4.1. The principally generated case

Proposition 4.1. *Suppose $f, g \in \mathbf{T}[\mathbf{x}]$ and $\mathbf{V}(f, g) = \emptyset$. Then there exists $h \in \mathbf{T}[\mathbf{x}]$ with nonzero constant term such that $(h, \epsilon h) \in \langle (f, g) \rangle$ for some real number $\epsilon > 0$.*

Proof. By Corollary 3.6, we may assume $f(\mathbf{a}) < g(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{T}^n$. (Note that this implies g has nonzero constant term.) By Corollary 3.8, there then exists $\epsilon > 0$ such that $\epsilon f(\mathbf{a}) < g(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{T}^n$. Notice that this implies $(g, g + \epsilon f) \in \mathbf{E}(\mathbf{T}^n)$. By Corollary 3.19, it then follows that $(g^m)_{\text{sat}} g^m = (g^m)_{\text{sat}} (g + \epsilon f)^m$ for m sufficiently large. Now observe that

$$(\epsilon f, g + \epsilon f) = (f, g) + (\epsilon f, \epsilon f) \in \langle (f, g) \rangle,$$

and hence

$$\begin{aligned} ((g^m)_{\text{sat}} (\epsilon f)^m, (g^m)_{\text{sat}} g^m) &= ((g^m)_{\text{sat}} (\epsilon f)^m, (g^m)_{\text{sat}} (g + \epsilon f)^m) \\ &= ((g^m)_{\text{sat}}, (g^m)_{\text{sat}}) \cdot (\epsilon f, g + \epsilon f)^m \in \langle (f, g) \rangle. \end{aligned}$$

On the other hand, we also have

$$((g^m)_{\text{sat}} (\epsilon f)^m, (g^m)_{\text{sat}} (\epsilon g)^m) = ((g^m)_{\text{sat}} \epsilon^m, (g^m)_{\text{sat}} \epsilon^m) \cdot (f, g)^m \in \langle (f, g) \rangle,$$

and so by transitivity $((g^m)_{\text{sat}} g^m, \epsilon^m (g^m)_{\text{sat}} g^m) \in \langle (f, g) \rangle$. The result now follows, taking $h = (g^m)_{\text{sat}} g^m$. $\square$

The epsilon that occurs in the above situation is not special:

Lemma 4.2. *Suppose E is a congruence on $\mathbf{T}[\mathbf{x}]$ and $h \in \mathbf{T}[\mathbf{x}]$. If $(h, \epsilon h) \in E$ for some real number $\epsilon > 0$, then $(h, \delta h) \in E$ for every $\delta \in \mathbf{R}$.*

Proof. First consider $\delta \geq 0$. Choose any N such that $\epsilon^N > \delta$ (in tropical operations). Notice that for every $n \geq 1$ one has

$$(\epsilon^{(n-1)}, \epsilon^{(n-1)}) \cdot (h, \epsilon h) = (\epsilon^{(n-1)} h, \epsilon^n h) \in E.$$

By transitivity and induction, it follows that $(h, \epsilon^n h) \in E$ for every $n \geq 1$. In particular, it follows that

$$(\delta h, \epsilon^N h) = (h, \epsilon^N h) + (\delta h, \delta h) \in E,$$

and so again by transitivity $(h, \delta h) \in E$. Now suppose $\delta < 0$. Then $\delta^{-1} > 0$ (again, in tropical operations), and so by the above argument $(h, \delta^{-1} h) \in E$. It immediately follows that $(h, \delta h) = (\delta, \delta) \cdot (\delta^{-1} h, h) \in E$. $\square$

4.2. Flatness

Proposition 4.1 suggests the possibility that $\mathbf{V}(E)$ is empty exactly when E contains a pair of the form $(h, \epsilon h)$, with $h \in \mathbf{T}[\mathbf{x}]$ having nonzero constant term and $\epsilon \neq \mathbf{1}_{\mathbf{T}}$. We therefore consider a new property of congruences on $\mathbf{T}[\mathbf{x}]$ called “flatness.” We begin by defining a related property that can be applied more generally to a congruence on any $\mathbf{T}$ -semialgebra. We say a congruence E on a $\mathbf{T}$ -semialgebra R is **flat over $\mathbf{T}$** if multiplication by $\mathbf{T}$ is a faithful action on R/E , i.e., if whenever $(sr, tr) \in E$ for some $s, t \in \mathbf{T}$ one has either $s = t$ or $(r, 0_R) \in E$. We say R is **flat over $\mathbf{T}$** if Δ_R is flat over $\mathbf{T}$. Flatness has many desirable properties. For instance, if R is flat over $\mathbf{T}$ then the structure morphism $i : \mathbf{T} \rightarrow R$ is injective.

Example 4.3. If R is a flat $\mathbf{T}$ -semialgebra, then $R[x]$ is also a flat $\mathbf{T}$ -semialgebra. Indeed, suppose $s \neq t \in \mathbf{T}$ and $f \in R[x]$ satisfies $sf = tf$. Writing $f(x) = \sum_d c_d x^d$, we must then have $\sum_d (sc_d)x^d = \sum_d (tc_d)x^d$, and hence $sc_d = tc_d$ for every d . As R is flat and $s \neq t$, this implies $c_d = 0_R$ for every d , and so $f = 0_{R[x]}$.

Example 4.4. By **Example 2.9**, the only congruences on $\mathbf{T}$ are $\Delta_{\mathbf{T}}$ and $E_{\mathbf{B}}$. The former is flat but the latter is not, as $\mathbf{T}/E_{\mathbf{B}} \cong \mathbf{B}$, which is not flat over $\mathbf{T}$.

Example 4.5. Suppose E is any congruence on $\mathbf{T}[\mathbf{x}]$, and suppose $(sh, th) \in \mathbf{E}(\mathbf{V}(E))$ for some $s, t \in \mathbf{T}$ and $h \in \mathbf{T}[\mathbf{x}]$. Then $s \cdot h(\mathbf{a}) = t \cdot h(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{V}(E)$, and so if $s \neq t$ then we must have $h(\mathbf{a}) = 0_{\mathbf{T}}$ for every $\mathbf{a} \in \mathbf{V}(E)$, i.e., $(h, 0_{\mathbf{T}}) \in \mathbf{E}(\mathbf{V}(E))$. Thus, $\mathbf{E}(\mathbf{V}(E))$ is always flat over $\mathbf{T}$.

For each $\mathbf{T}$ -semialgebra R , there is a canonical bijection between flat congruences on R and (isomorphism classes of) surjective $\mathbf{T}$ -semialgebra morphisms from R to flat $\mathbf{T}$ -semialgebras. By the usual Zorn’s lemma argument, every flat congruence on a $\mathbf{T}$ -semialgebra is contained in a maximal flat congruence. It is *not* the case that maximal flat congruences are necessarily maximal congruences, cf. **Example 2.9**. However, it is the case that maximal flat congruences are prime.

Proposition 4.6. *Suppose R is a flat $\mathbf{T}$ -semialgebra with no nontrivial flat congruences. Then R is an integral domain.*

Proof. Suppose not, so that there exists some nonzero $a \in R$ and $b \neq c \in R$ with $ab = ac$. Then the congruence $E_a = \{(d, e) \in R \times R \mid ad = ae\}$ is nontrivial. By assumption E_a cannot be flat, so there must exist $s \neq t \in \mathbf{T}$ and $d \in R$ with $(sd, td) \in E_a$ and $(d, 0_R) \notin E_a$, i.e., $a(sd) = a(td)$ and $ad \neq 0_R$. However, the first equality implies $s(ad) = t(ad)$. As R is flat and $s \neq t$, this implies $ad = 0_R$, a contradiction. $\square$

It follows that every maximal flat congruence on a $\mathbf{T}$ -semialgebra is prime, and hence every flat congruence is contained in a flat, prime congruence.

Example 4.7. We claim that for every $\mathbf{a} \in \mathbf{T}^n$ the congruence $\langle (x_1, a_1), \dots, (x_n, a_n) \rangle$ is a maximal flat congruence on $\mathbf{T}[\mathbf{x}]$. Indeed, following the argument in [Example 2.10](#), one can show $\langle (x_1, a_1), \dots, (x_n, a_n) \rangle$ is the kernel of the surjective semiring morphism $\text{ev}_{\mathbf{a}} : \mathbf{T}[\mathbf{x}] \rightarrow \mathbf{T}$. It follows that $\mathbf{T}[\mathbf{x}]/\langle (x_1, a_1), \dots, (x_n, a_n) \rangle = \mathbf{T}[\mathbf{x}]/\ker \text{ev}_{\mathbf{a}} \cong \mathbf{T}$, and hence $\langle (x_1, a_1), \dots, (x_n, a_n) \rangle$ is a maximal flat congruence on $\mathbf{T}[\mathbf{x}]$. (The converse of this statement is proven in [Corollary 4.16](#).)

We now restrict to the special case of the $\mathbf{T}$ -semialgebra $\mathbf{T}[\mathbf{x}]$. We say a congruence E on $\mathbf{T}[\mathbf{x}]$ is **weakly flat** if whenever $(h, \epsilon h) \in E$ for some $h \in \mathbf{T}[\mathbf{x}]$ and $\epsilon \neq 1_{\mathbf{T}}$, h necessarily has zero constant term. We will show that a congruence on $\mathbf{T}[\mathbf{x}]$ is weakly flat if and only if it is contained in a flat congruence. To prove this, we first need a few technical lemmas.

Lemma 4.8. *Suppose E is a congruence on $\mathbf{T}[\mathbf{x}]$ and $\phi_i : \mathbf{T}[\mathbf{x}] \rightarrow \mathbf{T}[x_1, \dots, \hat{x}_i, \dots, x_n]$ is the surjective semiring morphism given by setting $x_i = 0_{\mathbf{T}}$. Then $(\phi_i)_*(E)$ is not weakly flat if and only if there exist $h \in \mathbf{T}[x_1, \dots, \hat{x}_i, \dots, x_n]$, $A \in \mathbf{T}[\mathbf{x}]$ such that*

$$(1_{\mathbf{T}} + h + x_i A, \epsilon(1_{\mathbf{T}} + h) + x_i A) \in E$$

for every $\epsilon \in \mathbf{R}$.

Proof. The converse direction is clear. So suppose E is a congruence on $\mathbf{T}[\mathbf{x}]$ and $(\phi_i)_*(E)$ is not weakly flat, i.e., there exist $h \in \mathbf{T}[x_1, \dots, \hat{x}_i, \dots, x_n]$ and $\epsilon > 0$ such that

$$(1_{\mathbf{T}} + h, \epsilon(1_{\mathbf{T}} + h)) \in (\phi_i)_*(E).$$

(It immediately follows that the above holds for every $\epsilon \in \mathbf{R}$.) By [Corollary 2.14](#), this implies there exist $r_j \in \mathbf{T}[x_1, \dots, \hat{x}_i, \dots, x_n]$ and $f, g, a_j, b_j \in \mathbf{T}[\mathbf{x}]$ such that

$$(1_{\mathbf{T}} + h + x_i f, r_1 + x_i a_1), (r_1 + x_i b_1, r_2 + x_i a_2), \dots, \\ (r_m + x_i b_m, \epsilon(1_{\mathbf{T}} + h) + x_i g) \in E.$$

Adding $x_i(f + a_1 + b_1 + \dots + a_m + b_m + g)$ to both sides of every pair above yields a transitive chain in E . Transitivity in E then gives the result, with $A = f + a_1 + b_1 + \dots + a_m + b_m + g$. $\square$

Lemma 4.9. *Suppose E is a congruence on $\mathbf{T}[\mathbf{x}]$ and $\phi_i : \mathbf{T}[\mathbf{x}] \rightarrow \mathbf{T}[x_1, \dots, \hat{x}_i, \dots, x_n]$ is the surjective semiring morphism given by setting $x_i = 0_{\mathbf{T}}$. Then $(\phi_i)_*(E)$ is not weakly flat for all i if and only if for every monomial $\mathbf{m} \in \mathbf{T}[\mathbf{x}]$ there exist $h_{\mathbf{m}}, A_{\mathbf{m}} \in \mathbf{T}[\mathbf{x}]$ such that*

$$(1_{\mathbf{T}} + h_{\mathbf{m}} + \mathbf{m}A_{\mathbf{m}}, \epsilon(1_{\mathbf{T}} + h_{\mathbf{m}}) + \mathbf{m}A_{\mathbf{m}}) \in E$$

for every $\epsilon \in \mathbf{R}$.

Proof. The converse direction is clear, simply by considering the monomial $\mathbf{m} = x_i$ for each i . Now suppose E is a congruence on $\mathbf{T}[\mathbf{x}]$ such that for every i the congruence $(\phi_i)_*(E)$ is not weakly flat. By the previous lemma, for each i there exist $h_i \in \mathbf{T}[x_1, \dots, \hat{x}_i, \dots, x_n]$, $A_i \in \mathbf{T}[\mathbf{x}]$ such that

$$(1_{\mathbf{T}} + h_i + x_i A_i, \epsilon(1_{\mathbf{T}} + h_i) + x_i A_i) \in E$$

for every $\epsilon \in \mathbf{R}$. Multiplying both sides of this pair by $x_i A_i$ yields

$$(x_i A_i + x_i A_i h_i + x_i^2 A_i^2, \epsilon(x_i A_i + x_i A_i h_i) + x_i^2 A_i^2) \in E.$$

Adding this to the previous pair, we obtain

$$(1_{\mathbf{T}} + H_{i,2} + x_i^2 A_i^2, \epsilon(1_{\mathbf{T}} + H_{i,2}) + x_i^2 A_i^2) \in E,$$

where $H_{i,2} = h_i + x_i A_i + x_i A_i h_i$. Continuing this process inductively, it follows that for every d there exists $H_{i,d} \in \mathbf{T}[\mathbf{x}]$ such that

$$(1_{\mathbf{T}} + H_{i,d} + x_i^d A_i^d, \epsilon(1_{\mathbf{T}} + H_{i,d}) + x_i^d A_i^d) \in E.$$

Now suppose $\mathbf{m} \in \mathbf{T}[\mathbf{x}]$ is some monomial, say $\mathbf{m} = x_1^{d_1} \cdots x_n^{d_n}$. Let $A = A_1^{d_1} + \cdots + A_n^{d_n}$. Adding $x_i^{d_i} A$ to both sides of the above pair (with $d = d_i$) gives

$$(1_{\mathbf{T}} + H_{i,d_i} + x_i^{d_i} A, \epsilon(1_{\mathbf{T}} + H_{i,d_i}) + x_i^{d_i} A) \in E.$$

Consider for the moment this relation in the case $i = 1$. Multiplying both sides of that pair by $x_2^{d_2} A$ and adding the resulting pair to the pair for $i = 2$ gives

$$(1_{\mathbf{T}} + K_2 + x_1^{d_1} x_2^{d_2} A^2, \epsilon(1_{\mathbf{T}} + K_2) + x_1^{d_1} x_2^{d_2} A^2) \in E,$$

where $K_2 = H_{2,d_2} + x_2^{d_2} A + x_2^{d_2} A H_{1,d_1}$. Repeating this process, multiplying the above pair by $x_3^{d_3} A$ and adding the resulting pair to the pair for $i = 3$ gives

$$(1_{\mathbf{T}} + K_3 + x_1^{d_1} x_2^{d_2} x_3^{d_3} A^3, \epsilon(1_{\mathbf{T}} + K_3) + x_1^{d_1} x_2^{d_2} x_3^{d_3} A^3) \in E.$$

Continuing this inductive process, we ultimately obtain

$$(1_{\mathbf{T}} + K_n + \mathbf{m}A^n, \epsilon(1_{\mathbf{T}} + K_n) + \mathbf{m}A^n) \in E.$$

The result now follows, with $h_{\mathbf{m}} = K_n$ and $A_{\mathbf{m}} = A^n$. $\square$

We can now prove the following:

Proposition 4.10. *A congruence on $\mathbf{T}[\mathbf{x}]$ is weakly flat if and only if it is contained in a flat congruence on $\mathbf{T}[\mathbf{x}]$.*

Proof. First suppose E is a congruence on $\mathbf{T}[\mathbf{x}]$ contained in a flat congruence. Then E is contained in a maximal flat congruence, and hence in a flat, prime congruence P . Suppose $(h, \epsilon h) \in E$ for some $h \in \mathbf{T}[\mathbf{x}]$ and $\epsilon \neq 1_{\mathbf{T}}$. Then $(h, \epsilon h) \in P$, hence by primality and flatness $(h, 0_{\mathbf{T}}) \in P$. If a_0 is the constant term of h , then we also have $(a_0, 0_{\mathbf{T}}) \in P$ (simply by adding a_0 to both sides and using transitivity). Again by flatness, this implies $a_0 = 0_{\mathbf{T}}$. Thus, E is weakly flat.

For the converse direction we use induction on n , the number of variables. In the case $n = 0$ the notions of weakly flat and flat coincide, so there is nothing to prove. Now suppose the result holds in fewer than n variables for some $n > 0$, and suppose E is a weakly flat congruence on $\mathbf{T}[\mathbf{x}] = \mathbf{T}[x_1, \dots, x_n]$. If E is flat we are done. If E is not flat, then there exists nonzero $g \in \mathbf{T}[\mathbf{x}]$ with zero constant term such that $(g, \epsilon g) \in E$ for all $\epsilon \in \mathbf{R}$. Write $g = \sum_{\mathbf{m}} c_{\mathbf{m}} \mathbf{m}$, where the sum is over all monomials $\mathbf{m}$ that occur with nonzero coefficient. Rescaling if necessary, we may assume all $c_{\mathbf{m}} \geq 1_{\mathbf{T}}$. We claim that for some i the congruence $(\phi_i)_*(E)$ must be weakly flat. Suppose not. Then by [Lemma 4.9](#) for every monomial $\mathbf{m} \in \mathbf{T}[\mathbf{x}]$ there exist $h_{\mathbf{m}}, A_{\mathbf{m}} \in \mathbf{T}[\mathbf{x}]$ such that

$$(1_{\mathbf{T}} + h_{\mathbf{m}} + \mathbf{m}A_{\mathbf{m}}, \epsilon(1_{\mathbf{T}} + h_{\mathbf{m}}) + \mathbf{m}A_{\mathbf{m}}) \in E$$

for every $\epsilon \in \mathbf{R}$. Let $A = \sum A_{\mathbf{m}}$, where the sum runs over all monomials occurring in g with nonzero coefficient. For each monomial $\mathbf{m}$ occurring in g , add $c_{\mathbf{m}} \mathbf{m}A$ to both sides of the above pair to obtain

$$(1_{\mathbf{T}} + h_{\mathbf{m}} + c_{\mathbf{m}} \mathbf{m}A, \epsilon(1_{\mathbf{T}} + h_{\mathbf{m}}) + c_{\mathbf{m}} \mathbf{m}A) \in E.$$

Summing these pairs over all monomials occurring in g then gives

$$(1_{\mathbf{T}} + h + gA, \epsilon(1_{\mathbf{T}} + h) + gA) \in E,$$

where $h = \sum_{\mathbf{m}} h_{\mathbf{m}}$. Using the pair $(g, \epsilon g) \in E$ then gives

$$(1_{\mathbf{T}} + h + gA, \epsilon(1_{\mathbf{T}} + h + gA)) \in E.$$

But this violates the weak flatness of E . So, for some index i the congruence $(\phi_i)_*(E)$ must be weakly flat. By induction, it follows that $(\phi_i)_*(E)$ is contained in some flat congruence F_i on $\mathbf{T}[x_1, \dots, \hat{x}_i, \dots, x_n]$. The flat congruence F_i then corresponds to a flat congruence F on $\mathbf{T}[\mathbf{x}]$ that contains $\ker \phi_i$. Since $E \subseteq \ker \phi_i \subseteq F$, we've thus proven that E is contained in a flat congruence. $\square$

4.3. The general case

We now proceed to reduce the general case of a finitely generated congruence to the principally generated case. The main step is the following proposition.

Proposition 4.11. *Suppose E is a finitely generated congruence on $\mathbf{T}[\mathbf{x}]$ and $\mathbf{V}(E) \subseteq \mathbf{T}^n \setminus \mathbf{R}^n$. Then there exists a pair $(f, g) \in E$ with $\mathbf{V}(f, g) \subseteq \mathbf{T}^n \setminus \mathbf{R}^n$.*

To prove this proposition, we proceed in two steps. First we replace the given collection of generators with a transitive chain of generators that define the same congruence variety as E away from $\mathbf{T}^n \setminus \mathbf{R}^n$.

Lemma 4.12. *Suppose $E = \langle (f_1, g_1), \dots, (f_k, g_k) \rangle$ is a finitely generated congruence on $\mathbf{T}[\mathbf{x}]$. Then there exist pairs $(h_0, h_1), (h_1, h_2), \dots, (h_{k-1}, h_k) \in E$ such that*

$$\mathbf{V}((h_0, h_1), (h_1, h_2), \dots, (h_{k-1}, h_k)) \subseteq \mathbf{V}(E) \cup \bigcup_{i=1}^k \mathbf{V}(f_i, 0_{\mathbf{T}})$$

Proof. Define $h_0 = f_1 \cdots f_k$, and $h_i = g_i \prod_{j \neq i} f_j$, for each $i = 1, \dots, k$, where the product is over $1 \leq j \leq k$ with $j \neq i$. Then $(h_0, h_1) = (f_1, g_1) \prod_{j \geq 2} (f_j, f_j) \in E$, and for each $i = 1, \dots, k-1$, $(h_i, h_{i+1}) = (g_i, f_i) \cdot (f_{i+1}, g_{i+1}) \prod_{j \neq i, i+1} (f_j, f_j) \in E$, where the product is over $1 \leq j \leq k$ with $j \neq i, i+1$. Now suppose $\mathbf{a} \in \mathbf{V}((h_0, h_1), (h_1, h_2), \dots, (h_{k-1}, h_k))$, and suppose $f_j(\mathbf{a}) \neq 0_{\mathbf{T}}$ for every j . Then observe that for every i we have $h_0(\mathbf{a}) = h_i(\mathbf{a})$, which is the equality

$$f_1(\mathbf{a}) \cdots f_i(\mathbf{a}) \cdots f_k(\mathbf{a}) = f_1(\mathbf{a}) \cdots g_i(\mathbf{a}) \cdots f_k(\mathbf{a}).$$

As we're assuming all the $f_j(\mathbf{a})$ are nonzero, this equality implies $f_i(\mathbf{a}) = g_i(\mathbf{a})$. As this is true for every i , we have $\mathbf{a} \in \mathbf{V}(E)$. $\square$

We next replace this transitive chain of pairs with a single pair having the same congruence variety.

Lemma 4.13. *Given $h_0, \dots, h_k \in \mathbf{T}[\mathbf{x}]$, we always have:*

- i) $((h_0 + \cdots + h_k)^{k+1}, h_0 \cdots h_k) \in \langle (h_0, h_1), (h_1, h_2), \dots, (h_{k-1}, h_k) \rangle$; and
- ii) $\mathbf{V}((h_0, h_1), (h_1, h_2), \dots, (h_{k-1}, h_k)) = \mathbf{V}((h_0 + \cdots + h_k)^{k+1}, h_0 \cdots h_k)$.

Proof. Let $E = \langle (h_0, h_1), (h_1, h_2), \dots, (h_{k-1}, h_k) \rangle$. First observe that, whenever $i_0, \dots, i_k$ are nonnegative integers that sum to $k+1$, we have

$$(h_0^{i_0} \cdots h_k^{i_k}, h_0 \cdots h_k) = \prod_{j=0}^k \prod_{l=1}^{i_l} (h_j, h_l) \in E.$$

As the addition in $\mathbf{T}[\mathbf{x}]$ is idempotent, it follows that

$$\begin{aligned} ((h_0 + \cdots + h_k)^{k+1}, h_0 \cdots h_k) &= \left(\sum_{i_0 + \cdots + i_k = k+1} h_0^{i_0} \cdots h_k^{i_k}, h_0 \cdots h_k \right) \\ &= \sum_{i_0 + \cdots + i_k = k+1} (h_0^{i_0} \cdots h_k^{i_k}, h_0 \cdots h_k) \in E. \end{aligned}$$

Thus,

$$\mathbf{V}(E) \subseteq \mathbf{V}((h_0 + \cdots + h_k)^{k+1}, h_0 \cdots h_k).$$

Now suppose $\mathbf{a} \in \mathbf{V}((h_0 + \cdots + h_k)^{k+1}, h_0 \cdots h_k)$. Then (in classical operations)

$$(k+1) \cdot \max\{h_0(\mathbf{a}), \dots, h_k(\mathbf{a})\} = h_0(\mathbf{a}) + \cdots + h_k(\mathbf{a}),$$

which is only possible if $h_0(\mathbf{a}) = \cdots = h_k(\mathbf{a})$. Thus, $\mathbf{a} \in \mathbf{V}(E)$. $\square$

We can now prove [Proposition 4.11](#).

Proof of Proposition 4.11. Apply [Lemmas 4.12 and 4.13](#), together with the observation that one always has $\mathbf{V}(f, 0_{\mathbf{T}}) \subseteq \mathbf{T}^n \setminus \mathbf{R}^n$ for any $f \in \mathbf{T}[\mathbf{x}]$ (as all tropical functions are real-valued on $\mathbf{R}^n$). $\square$

Corollary 4.14. *Suppose E is a finitely generated congruence on $\mathbf{T}[\mathbf{x}]$ and $\mathbf{V}(E) \subseteq \mathbf{T}^n \setminus \mathbf{R}^n$. Then $(h, \epsilon h) \in E$ for some nonzero $h \in \mathbf{T}[\mathbf{x}]$ and real number $\epsilon > 0$.*

Proof. By [Proposition 4.11](#) there exists $(f, g) \in E$ with $\mathbf{V}(f, g) \subseteq \mathbf{T}^n \setminus \mathbf{R}^n$. In view of [Lemma 3.5](#) we may assume without loss of generality that $f(\mathbf{a}) \leq g(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{T}^n$ (with strict inequality in $\mathbf{R}^n$). By [Lemma 3.7](#) there then exists some $\epsilon > 0$ such that (in tropical operations) $\epsilon f(\mathbf{a}) \leq g(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{T}^n$, with equality only when $f(\mathbf{a}) = 0_{\mathbf{T}} = g(\mathbf{a})$. In particular, it follows that $(g + \epsilon f, g) \in \mathbf{E}(\mathbf{T}^n)$. By [Corollary 3.19](#) it follows that there exists $m \geq 1$ such that $(g^m)_{\text{sat}}(g + \epsilon f)^m = (g^m)_{\text{sat}}g^m$. Next observe that $(g + \epsilon f, \epsilon g) = (g, g) + (\epsilon f, \epsilon g) \in E$, hence $((g^m)_{\text{sat}}(g + \epsilon f)^m, (g^m)_{\text{sat}}(\epsilon g)^m) \in E$. Using the previous equality, this implies $((g^m)_{\text{sat}}g^m, (g^m)_{\text{sat}}(\epsilon g)^m) \in E$, and hence $((g^{2m})_{\text{sat}}, \epsilon^m (g^{2m})_{\text{sat}}) \in E$. $\square$

We can now prove the following weak form of the Tropical Nullstellensatz for Congruences:

Theorem 2. *Suppose E is a finitely generated congruence on $\mathbf{T}[\mathbf{x}]$. Then $\mathbf{V}(E)$ is empty if and only if E is not weakly flat, i.e., there exists $h \in \mathbf{T}[\mathbf{x}]$ with nonzero constant term such that $(h, \epsilon h) \in E$ for some $\epsilon \neq 1_{\mathbf{T}}$ (and hence all $\epsilon \in \mathbf{R}$).*

Proof. By Proposition 4.10, it is equivalent to prove $\mathbf{V}(E)$ is empty if and only if E is not contained in any flat congruence. We proceed by induction on the n , the number of variables. We first consider the case $n = 0$, where $\mathbf{T}^0$ is just a point and $\mathbf{T}[\emptyset] = \mathbf{T}$. By Example 2.9, the semiring $\mathbf{T}$ only has three congruences: the trivial congruence $\Delta_{\mathbf{T}}$, the maximal congruence $E_{\mathbf{B}}$, and the improper congruence $E_{\mathbf{T}}$. The only congruences for which $\mathbf{V}(E)$ is empty are $E_{\mathbf{B}}$ and $E_{\mathbf{T}}$. As $E_{\mathbf{B}}$ is maximal and not flat, it is not contained in any flat congruence. As $E_{\mathbf{T}}$ is improper, it is obviously not contained in any flat congruence. Thus, the statement holds for $n = 0$.

Now assume the statement holds when there are n or fewer variables, for some $n \geq 0$. Let E be a finitely generated congruence on $\mathbf{T}[x_1, \dots, x_{n+1}]$. It is clear that if E is not weakly flat then $\mathbf{V}(E)$ is empty. Conversely, suppose $\mathbf{V}(E)$ is empty but suppose, towards a contradiction, that E is weakly flat, and hence contained in a flat congruence. Then E is contained in a flat, prime congruence P . By Corollary 4.14 there exist nonzero $h \in \mathbf{T}[x_1, \dots, x_{n+1}]$ and $\epsilon > 1_{\mathbf{T}}$ such that $(h, \epsilon h) \in E$. Then $(h, \epsilon h) \in P$, so by primality and flatness we have $(h, 0_{\mathbf{T}}) \in P$. Let $a_{\mathbf{d}}$ be a nonzero coefficient of a monomial $\mathbf{x}^{\mathbf{d}}$ in h . As tropical addition is idempotent, it follows that $h(\mathbf{x}) + a_{\mathbf{d}}\mathbf{x}^{\mathbf{d}} = h(\mathbf{x})$, and hence $(a_{\mathbf{d}}\mathbf{x}^{\mathbf{d}}, 0_{\mathbf{T}}) \in P$. Again by primality and flatness, this implies $(x_i, 0_{\mathbf{T}}) \in P$ for some index $i \in \{1, \dots, n+1\}$. Relabeling if necessary, we may assume $i = n+1$. Consider now the surjective semiring morphism $\phi : \mathbf{T}[x_1, \dots, x_{n+1}] \rightarrow \mathbf{T}[x_1, \dots, x_n]$ given by setting $x_{n+1} = 0_{\mathbf{T}}$. By Corollary 2.15, if $E = \langle (f_1, g_1), \dots, (f_k, g_k) \rangle$, then $\phi_*(E) = \langle f_1(\mathbf{x}, 0_{\mathbf{T}}), g_1(\mathbf{x}, 0_{\mathbf{T}}), \dots, (f_k(\mathbf{x}, 0_{\mathbf{T}}), g_k(\mathbf{x}, 0_{\mathbf{T}})) \rangle$. In particular, $\mathbf{V}_{\mathbf{T}^n}(\phi_*(E)) \cong \mathbf{V}_{\mathbf{T}^{n+1}}(E) \cap \mathbf{V}_{\mathbf{T}^{n+1}}((x_{n+1}, 0_{\mathbf{T}})) = \emptyset$. So, $\phi_*(E)$ is a finitely-generated congruence on $\mathbf{T}[x_1, \dots, x_n]$ with $\mathbf{V}(\phi_*(E)) = \emptyset$. By the induction hypothesis it follows that $\phi_*(E)$ cannot be contained in any flat, prime congruence. However, we certainly have $\phi_*(E) \subseteq \phi_*(P)$, and we claim $\phi_*(P)$ is a flat, prime congruence. Indeed, as $(x_{n+1}, 0_{\mathbf{T}}) \in P$ one can verify $\mathbf{T}[x_1, \dots, x_n]/\phi_*(P) \cong \mathbf{T}[x_1, \dots, x_{n+1}]/P$. As P is flat and prime, the $\mathbf{T}$ -semialgebra $\mathbf{T}[x_1, \dots, x_{n+1}]/P$ is a flat integral domain, hence so is $\mathbf{T}[x_1, \dots, x_n]/\phi_*(P)$, i.e., $\phi_*(P)$ is also flat and prime. This is a contradiction. $\square$

Corollary 4.15. *If E is a finitely generated flat congruence on $\mathbf{T}[\mathbf{x}]$, then $\mathbf{V}(E)$ is nonempty.*

Corollary 4.16. *The finitely generated maximal flat congruences on $\mathbf{T}[\mathbf{x}]$ are precisely the congruences of the form $\langle (x_1, a_1), \dots, (x_n, a_n) \rangle$, for $\mathbf{a} \in \mathbf{T}^n$.*

Proof. The congruence $\langle (x_1, a_1), \dots, (x_n, a_n) \rangle$ is a maximal flat congruence on $\mathbf{T}[\mathbf{x}]$ by [Example 4.7](#). Also note that $\mathbf{V}(\langle (x_1, a_1), \dots, (x_n, a_n) \rangle) = \{\mathbf{a}\}$ and $\mathbf{E}(\mathbf{V}(\langle (x_1, a_1), \dots, (x_n, a_n) \rangle)) = \langle (x_1, a_1), \dots, (x_n, a_n) \rangle$, the latter by [Example 4.5](#) and the maximality of $\langle (x_1, a_1), \dots, (x_n, a_n) \rangle$. Now suppose E is any finitely generated flat congruence on $\mathbf{T}[\mathbf{x}]$. By the previous corollary, there is some point $\mathbf{a} \in \mathbf{V}(E)$. It follows that

$$E \subseteq \mathbf{E}(\mathbf{V}(E)) \subseteq \mathbf{E}(\{\mathbf{a}\}) = \mathbf{E}(\mathbf{V}(\langle (x_1, a_1), \dots, (x_n, a_n) \rangle)) = \langle (x_1, a_1), \dots, (x_n, a_n) \rangle.$$

In particular, if E is a maximal flat congruence, then we must have $E = \langle (x_1, a_1), \dots, (x_n, a_n) \rangle$. $\square$

5. The strong tropical Nullstellensatz for congruences

We now aim to determine the structure of $\mathbf{E}(\mathbf{V}(E))$. We begin by considering a reasonable candidate for $\mathbf{E}(\mathbf{V}(E))$.

5.1. A candidate for $\mathbf{E}(\mathbf{V}(E))$

5.1.1. The radical of a congruence

Classically, the Nullstellensatz states that $\mathbf{I}(\mathbf{V}(I)) = \sqrt{I}$, i.e., $f \in \mathbf{I}(\mathbf{V}(I))$ iff $f^n \in I$ for some $n \geq 1$. Similarly, in the tropical situation it is clear that if $(f^n, g^n) \in E$ for some $n \geq 1$, then $(f, g) \in \mathbf{E}(\mathbf{V}(E))$. The same holds true for twisted powers: if $(f, g)^{\times n} \in E$ for some $n \geq 1$, then $(f, g) \in \mathbf{E}(\mathbf{V}(E))$. These two properties can be combined with the following easily-proven fact: if either $(f^n, g^n) \in E$ or $(f, g)^{\times n} \in E$ for some $n \geq 1$, then $((f+g)^N, 0_{\mathbf{T}}) \times (f, g) \in E$ for sufficiently large N . Note that if $((f+g)^N, 0_{\mathbf{T}}) \times (f, g) \in E$, then $(f, g) \in \mathbf{E}(\mathbf{V}(E))$.

There is another issue to consider. Recall [Lemma 3.1](#), which states that $\mathbf{V}((h, k) \times (f, g)) = \mathbf{V}(h, k) \cup \mathbf{V}(f, g)$. It follows that if $\mathbf{V}(h, k) = \emptyset$, then $\mathbf{V}((h, k) \times (f, g)) = \mathbf{V}(f, g)$. In particular, if $(h, k) \times (f, g) \in E$ and $\mathbf{V}(h, k) = \emptyset$, then $(f, g) \in \mathbf{E}(\mathbf{V}(E))$. By [Theorem 2](#), a congruence has an empty associated congruence variety exactly when the congruence contains a pair of the form $(h, \epsilon h) = (1_{\mathbf{T}}, \epsilon) \times (h, 0_{\mathbf{T}})$, where $h \in \mathbf{T}[\mathbf{x}]$ has nonzero constant term and $\epsilon \neq 1_{\mathbf{T}}$. Rescaling if necessary, we may assume the constant term of h is $1_{\mathbf{T}}$, and so h is of the form $h = 1_{\mathbf{T}} + r$ for some $r \in \mathbf{T}[\mathbf{x}]$. In particular, we see that if $(1_{\mathbf{T}}, \epsilon) \times (1_{\mathbf{T}} + r, 0_{\mathbf{T}}) \times (f, g) \in E$ for some $r \in \mathbf{T}[\mathbf{x}]$ and $\epsilon \neq 1_{\mathbf{T}}$, then $(f, g) \in \mathbf{E}(\mathbf{V}(E))$.

Combining the previous two observations, to any congruence E on $\mathbf{T}[\mathbf{x}]$ we consider the collection of all pairs (f, g) for which the following condition holds: there exist $r \in \mathbf{T}[\mathbf{x}]$, $\epsilon \neq 1_{\mathbf{T}}$, and $N > 0$ such that

$$(1_{\mathbf{T}}, \epsilon) \times ((f+g)^N + r, 0_{\mathbf{T}}) \times (f, g) \in E.$$

We call this collection the **pre-radical** of E and denote it $\text{rad}^-(E)$. By construction, we have $E \subseteq \text{rad}^-(E) \subseteq \mathbf{E}(\mathbf{V}(E))$. While it is not obvious the set $\text{rad}^-(E)$ is a congruence (but see [Remark 5.6](#)), it does have several nice properties.

Lemma 5.1. *If $(f, g) \in \text{rad}^-(E)$, then there exist $r \in \mathbf{T}[\mathbf{x}]$ and $N > 0$ such that $(1_{\mathbf{T}}, \delta) \rtimes ((f + g)^N + r, 0_{\mathbf{T}}) \rtimes (f, g) \in E$ for every $\delta \in \mathbf{R}$.*

Proof. Suppose $(f, g) \in \text{rad}^-(E)$. Then there exist $r \in \mathbf{T}[\mathbf{x}]$, $\epsilon \neq 1_{\mathbf{T}}$, and $N > 0$ such that

$$(1_{\mathbf{T}}, \epsilon) \rtimes ((f + g)^N + r, 0_{\mathbf{T}}) \rtimes (f, g) \in E. \quad (*)$$

If $\epsilon = 0_{\mathbf{T}}$, then $(*)$ states $((f + g)^N + r, 0_{\mathbf{T}}) \rtimes (f, g) \in E$. Twisted multiplying by $(1_{\mathbf{T}}, \delta)$ then gives the desired result. Now suppose $\epsilon \neq 0_{\mathbf{T}}$. Observe that twisted multiplying $(*)$ by $(0_{\mathbf{T}}, \epsilon^{-1})$ implies

$$(1, \epsilon^{-1}) \rtimes ((f + g)^N + r, 0_{\mathbf{T}}) \rtimes (f, g) \in E,$$

and so $(*)$ holds with ϵ replaced by ϵ^{-1} . We may therefore assume $\epsilon > 1_{\mathbf{T}}$. We claim $(*)$ also holds when ϵ is replaced by ϵ^2 . Indeed, let $k = (f + g)^N + r$, and observe that (since $\epsilon^2 > \epsilon > 1_{\mathbf{T}}$) we have

$$\begin{aligned} k(f + \epsilon^2 g) &= k(f + \epsilon g + \epsilon^2 g) \sim_E k(\epsilon f + g + \epsilon^2 g) = k(\epsilon f + \epsilon^2 g) = \epsilon k(f + \epsilon g) \\ &\sim_E \epsilon k(\epsilon f + g) = k(\epsilon^2 f + \epsilon g) = k(\epsilon^2 f + f + \epsilon g) \\ &\sim_E h(\epsilon^2 f + \epsilon f + g) = k(\epsilon^2 f + g). \end{aligned}$$

(Here we are using the notation $a \sim_E b$ to indicate $(a, b) \in E$.) Thus, we indeed have $(1, \epsilon^2) \rtimes ((f + g)^N + r, 0_{\mathbf{T}}) \rtimes (f, g) \in E$. Proceeding inductively, one can then easily show $(*)$ holds with ϵ replaced by ϵ^M for any $M \geq 1$ (without changing the constant N).

Now take any $\delta \in \mathbf{R}$. Replacing ϵ with ϵ^M for M sufficiently large, we may assume $\epsilon^{-1} \leq \delta \leq \epsilon$. Observe that

$$(\epsilon^{-1} \delta, \epsilon^{-1}) \rtimes (1_{\mathbf{T}}, \epsilon) = ((\epsilon^{-1} \delta) + 1_{\mathbf{T}}, \delta + \epsilon^{-1}) = (1_{\mathbf{T}}, \delta),$$

and so twisted multiplying $(*)$ by $(\epsilon^{-1} \delta, \epsilon^{-1})$ implies

$$(1, \delta) \rtimes ((f + g)^N + r, 0_{\mathbf{T}}) \rtimes (f, g) \in E. \quad \square$$

Lemma 5.2. *For any congruence E on $\mathbf{T}[\mathbf{x}]$, we have $\mathbf{V}(\text{rad}^-(E)) = \mathbf{V}(E)$.*

Proof. Since $E \subseteq \text{rad}^-(E) \subseteq \mathbf{E}(\mathbf{V}(E))$, it follows that $\mathbf{V}(E) = \mathbf{V}(\mathbf{E}(\mathbf{V}(E))) \subseteq \mathbf{V}(\text{rad}^-(E)) \subseteq \mathbf{V}(E)$, and hence $\mathbf{V}(\text{rad}^-(E)) = \mathbf{V}(E)$. $\square$

Example 5.3. By definition and the above lemma, $(1_{\mathbf{T}}, 0_{\mathbf{T}}) \in \text{rad}^-(E)$ if and only if there exist $r \in \mathbf{T}[\mathbf{x}]$ and $N > 0$ such that $(1_{\mathbf{T}}, \epsilon) \rtimes ((1_{\mathbf{T}} + 0_{\mathbf{T}})^N + r, 0_{\mathbf{T}}) \rtimes (1_{\mathbf{T}}, 0_{\mathbf{T}}) \in E$ for some $\epsilon \neq 1_{\mathbf{T}}$ (equivalently, all $\epsilon \in \mathbf{R}$). Simplifying, we see that $(1_{\mathbf{T}}, 0_{\mathbf{T}}) \in \text{rad}^-(E)$ if and only if there exists $r \in \mathbf{T}[\mathbf{x}]$ such that $(1_{\mathbf{T}}, \epsilon) \rtimes (1_{\mathbf{T}} + r, 0_{\mathbf{T}}) \in E$ for some $\epsilon \neq 1_{\mathbf{T}}$ (equivalently, all $\epsilon \in \mathbf{R}$), i.e., if and only if E is not weakly flat.

In light of the above example, we see that in this new terminology we can restate [Theorem 2](#) as follows:

Theorem 2. *If E is a finitely generated congruence on $\mathbf{T}[\mathbf{x}]$, then $\mathbf{V}(E)$ is empty if and only if $(1_{\mathbf{T}}, 0_{\mathbf{T}}) \in \text{rad}^-(E)$.*

Example 5.4. Let E be a congruence on $\mathbf{T}[\mathbf{x}]$ and $f \in \mathbf{T}[\mathbf{x}]$ be a fixed polynomial. If $(f, 0_{\mathbf{T}}) \in \text{rad}^-(E)$, then there exist $r \in \mathbf{T}[\mathbf{x}]$ and $N > 0$ such that $(1_{\mathbf{T}}, \epsilon) \times (f^N + r, 0_{\mathbf{T}}) \times (f, 0_{\mathbf{T}}) \in E$ for some $\epsilon \neq 1_{\mathbf{T}}$. Taking $r' = rf$ and $N' = N + 1$, it follows that $(1_{\mathbf{T}}, \epsilon) \times (f^{N'} + r', 0_{\mathbf{T}}) \in \text{rad}^-(E)$. Conversely, if there exist $r' \in \mathbf{T}[\mathbf{x}]$ and $N' > 0$ such that $(1_{\mathbf{T}}, \epsilon) \times (f^{N'} + r', 0_{\mathbf{T}}) \in \text{rad}^-(E)$ for some $\epsilon \neq 1_{\mathbf{T}}$, then twisted multiplying by $(f, 0_{\mathbf{T}})$ implies $(1_{\mathbf{T}}, \epsilon) \times (f^{N'} + r', 0_{\mathbf{T}}) \times (f, 0_{\mathbf{T}}) \in E$, and hence $(f, 0_{\mathbf{T}}) \in \text{rad}^-(E)$. Thus, $(f, 0_{\mathbf{T}}) \in \text{rad}^-(E)$ if and only if there exist $r \in \mathbf{T}[\mathbf{x}]$ and $N > 0$ such that $(1_{\mathbf{T}}, \epsilon) \times (f^N + r, 0_{\mathbf{T}}) \in E$ for some $\epsilon \neq 1_{\mathbf{T}}$ (equivalently, all $\epsilon \neq 0_{\mathbf{T}}$).

As usual, because of transitivity concerns there is a slightly more complicated situation to consider. Let $\text{rad}(E)$ denote the set of all pairs (f, g) for which there exist $k_0, \dots, k_{l+1} \in \mathbf{T}[\mathbf{x}]$ such that for each $i = 0, \dots, l$ one has $(k_i, k_{i+1}) \times (f, g) \in E$, and where $k_0 = (f + g)^N + r$ and $k_{l+1} = \epsilon k_0$ for some $r \in \mathbf{T}[\mathbf{x}]$, $\epsilon \neq 1_{\mathbf{T}}$, and $N > 0$. We call $\text{rad}(E)$ the **radical of E** .

While it is once again not obvious $\text{rad}(E)$ is a congruence, we certainly have $E \subseteq \text{rad}^-(E) \subseteq \text{rad}(E)$. We claim we also have $\text{rad}(E) \subseteq \mathbf{E}(\mathbf{V}(E))$. Indeed, suppose $(f, g) \in \text{rad}(E)$ but suppose, towards a contradiction, $(f, g) \notin \mathbf{E}(\mathbf{V}(E))$. Then there exists $\mathbf{a} \in \mathbf{V}(E)$ at which $f(\mathbf{a}) \neq g(\mathbf{a})$. From the pair $(k_i, k_{i+1}) \times (f, g) \in E$ it follows that we must then have $k_i(\mathbf{a}) = k_{i+1}(\mathbf{a})$ for each $i = 0, \dots, l$, and hence $k_0(\mathbf{a}) = k_{l+1}(\mathbf{a})$. As $k_{l+1} = \epsilon k_0$ with $\epsilon \neq 1_{\mathbf{T}}$, this implies $k_0(\mathbf{a}) = 0_{\mathbf{T}}$. But $k_0(\mathbf{a}) = (f(\mathbf{a}) + g(\mathbf{a}))^N + r(\mathbf{a})$, so $k_0(\mathbf{a}) = 0_{\mathbf{T}}$ only if $f(\mathbf{a}) = g(\mathbf{a}) = 0_{\mathbf{T}}$, a contradiction.

Lemma 5.5. *For any congruence E on $\mathbf{T}[\mathbf{x}]$, we have $\mathbf{V}(E) = \mathbf{V}(\text{rad}(E))$.*

Proof. Since $E \subseteq \text{rad}(E) \subseteq \mathbf{E}(\mathbf{V}(E))$, it follows that $\mathbf{V}(E) = \mathbf{V}(\mathbf{E}(\mathbf{V}(E))) \subseteq \mathbf{V}(\text{rad}(E)) \subseteq \mathbf{V}(E)$, and hence $\mathbf{V}(\text{rad}(E)) = \mathbf{V}(E)$. $\square$

Remark 5.6. The condition defining $\text{rad}^-(E)$ was recently studied more extensively in [\[34\]](#). When working in an additively idempotent semiring R they define the *generalized power* of a pair $(a, b) \in R \times R$ as any element of the form $((a + b)^k + c, 0) \times (a, b)^{\times l}$, where k, l are nonnegative integers and $c \in R$. They then show that the intersection of all twisted-prime congruences that contain E equals the collection of all pairs that have a generalized power contained in E ; see [\[34, Theorem 3.9\]](#). Additionally, they are able to use this result to prove that the set $\text{rad}^-(E)$ is in fact always a congruence [\[34, Prop. 5.1ii\]](#); moreover, when E is finitely generated one also has $\text{rad}^-(E) = \text{rad}(E) = \mathbf{E}(\mathbf{V}(E))$ [\[34, Prop. 5.3ii\]](#).

5.1.2. Limit pairs

We begin with the following motivating example:

Example 5.7. Suppose E is a congruence on $\mathbf{T}[\mathbf{x}]$ and $f \in \mathbf{T}[\mathbf{x}]$ is a tropical polynomial such that $(f + t, t) \in \mathbf{E}(\mathbf{V}(E))$ for every $t > 1_{\mathbf{T}}$, i.e., such that $f(\mathbf{a}) \leq t$ for every $\mathbf{a} \in \mathbf{V}(E)$ and $t > 1_{\mathbf{T}}$. It follows that $f(\mathbf{a}) \leq 1_{\mathbf{T}}$ for every $\mathbf{a} \in \mathbf{V}(E)$, and hence $(f + 1_{\mathbf{T}}, 1_{\mathbf{T}}) \in \mathbf{E}(\mathbf{V}(E))$. Similarly, if $(f + s, f) \in \mathbf{E}(\mathbf{V}(E))$ for every $s < 1_{\mathbf{T}}$, then $f(\mathbf{a}) \geq s$ for every $\mathbf{a} \in \mathbf{V}(E)$ and $s < 1_{\mathbf{T}}$, hence $f(\mathbf{a}) \geq 1_{\mathbf{T}}$ for every $\mathbf{a} \in \mathbf{V}(E)$, and hence $(f + 1_{\mathbf{T}}, f) \in \mathbf{E}(\mathbf{V}(E))$. If both of the aforementioned properties hold, then it follows by transitivity that $(f, 1_{\mathbf{T}}) \in \mathbf{E}(\mathbf{V}(E))$.

The above example suggests $\mathbf{E}(\mathbf{V}(E))$ is somehow “closed under limits,” a property not shared by arbitrary congruences. (Indeed, the congruence $F = \bigcup_{t > 1_{\mathbf{T}}} \langle (x + t, x) \rangle$ does not possess this property.) Let $\{f_i\}_i$ be a sequence of tropical polynomials, and write $f_i(\mathbf{x}) = \sum_{\mathbf{d}} c_{\mathbf{d}}^{(i)} \mathbf{x}^{\mathbf{d}}$. We say this sequence **converges** to a tropical polynomial $f(\mathbf{x}) = \sum_{\mathbf{d}} c_{\mathbf{d}} \mathbf{x}^{\mathbf{d}}$ if in each degree $\mathbf{d}$ the sequence of coefficients $\{c_{\mathbf{d}}^{(i)}\}_i$ converges to $c_{\mathbf{d}}$ (in the usual sense of convergence for extended real numbers). We write $\lim_i f_i = f$ to denote such a convergence. Since real limits commute with maximum and addition, these tropical limits commute with tropical addition and multiplication. Also, if $\lim_i f_i = f$, then $\lim_i f_i(\mathbf{a}) = f(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{T}[\mathbf{x}]$. We say a sequence of pairs $\{(f_i, g_i)\}_i$ **converges** to a pair (f, g) if $\lim_i f_i = f$ and $\lim_i g_i = g$. Given a subset E of $\mathbf{T}[\mathbf{x}] \times \mathbf{T}[\mathbf{x}]$, we say a pair (f, g) is a **limit pair** of E if it is the limit of a sequence of pairs $\{(f_i, g_i)\}_i$ contained in E . We write $\lim_i (f_i, g_i) = (f, g)$.

We would like to claim that if $\lim_i (f_i, g_i) = (f, g)$, then $\lim_i \mathbf{V}(f_i, g_i) = \mathbf{V}(f, g)$. However, it is not clear in what sense this latter limit is to be understood to converge, as we haven’t specified a topology on $\mathbf{T}^n$ (and points in $\mathbf{T}^n$ can be infinitely far away from each other). There is one scenario, however, in which this limit can be easily defined. We say a sequence of pairs $\{(f_i, g_i)\}_i$ is **ascending** if $\langle (f_i, g_i) \rangle \subseteq \langle (f_{i+1}, g_{i+1}) \rangle$ for every i . If $\{(f_i, g_i)\}_i$ is an ascending sequence, then $\mathbf{V}(f_{i+1}, g_{i+1}) \subseteq \mathbf{V}(f_i, g_i)$ for every i , and so we define $\lim_i \mathbf{V}(f_i, g_i) := \bigcap_i \mathbf{V}(f_i, g_i)$. We say a pair (f, g) is an **ascending limit** if it is a limit pair of an ascending sequence. Note that if (f, g) is the limit of an ascending sequence $\{(f_i, g_i)\}_i$, then $\mathbf{V}(f, g) = \lim_i \mathbf{V}(f_i, g_i)$. Denote by E^{lim} the collection of ascending limits of E .

Lemma 5.8. *For any subset $\mathcal{S} \subseteq \mathbf{T}[\mathbf{x}] \times \mathbf{T}[\mathbf{x}]$, we have $\mathbf{V}(\mathcal{S}) = \mathbf{V}(\mathcal{S}^{\text{lim}})$.*

Proof. As $\mathcal{S} \subseteq \mathcal{S}^{\text{lim}}$, we certainly have $\mathbf{V}(\mathcal{S}^{\text{lim}}) \subseteq \mathbf{V}(\mathcal{S})$. On the other hand, suppose $(f, g) \in \mathcal{S}^{\text{lim}}$. Then (f, g) is the limit of an ascending sequence $\{(f_i, g_i)\}_i$ in $\mathcal{S}$. Since for each i we have $(f_i, g_i) \in \mathcal{S}$, for each i we also have $\mathbf{V}(\mathcal{S}) \subseteq \mathbf{V}(f_i, g_i)$. It follows that $\mathbf{V}(\mathcal{S}) \subseteq \bigcap_i \mathbf{V}(f_i, g_i) = \mathbf{V}(f, g)$. As this is true for all $(f, g) \in \mathcal{S}^{\text{lim}}$, it follows that $\mathbf{V}(\mathcal{S}) \subseteq \bigcap_{(f, g) \in \mathcal{S}^{\text{lim}}} \mathbf{V}(f, g) = \mathbf{V}(\mathcal{S}^{\text{lim}})$. $\square$

Remark 5.9. Our need to consider ascending limits may stem from the fact that $\mathbf{T}[\mathbf{x}]$ is non-Noetherian. Indeed, it has recently been shown in [34] that when E is finitely generated there is no need to consider limit pairs.

We are now led to make the following definition. Given a congruence E on $\mathbf{T}[\mathbf{x}]$, let $\hat{E}$ denote the “one-link transitive closure” of $(\text{rad}(E))^{\text{lim}}$. In other words, a pair (f, g) is contained in $\hat{E}$ if and only if there exist ascending sequences $\{(f_i, h_i)\}_i, \{(h'_i, g_i)\}_i$ contained in $\text{rad}(E)$ with $\lim_i f_i = f, \lim_i h_i = \lim_i h'_i$, and $\lim_i g_i = g$.

By construction, for every congruence E on $\mathbf{T}[\mathbf{x}]$ we have $E \subseteq \text{rad}(E) \subseteq (\text{rad}(E))^{\text{lim}} \subseteq \hat{E}$. We also have $\mathbf{V}(E) = \mathbf{V}(\text{rad}(E)) = \mathbf{V}((\text{rad}(E))^{\text{lim}}) = \mathbf{V}(\hat{E})$, where the first equality holds by Lemma 5.5, the second by Lemma 5.8, and the third by construction. It follows that $\hat{E} \subseteq \mathbf{E}(\mathbf{V}(\hat{E})) = \mathbf{E}(\mathbf{V}(E))$. Our goal for the remainder of this paper is to prove the reverse containment. (In particular, this will also prove $\hat{E}$ is a congruence.) We proceed in three steps: first, by considering pairs in $\mathbf{E}(\mathbf{V}(E))$ of the form $(f, 0_{\mathbf{T}})$; second, by considering pairs of the form $(f, 1_{\mathbf{T}})$; and finally, by reducing the general case to the two previous cases. We will use a version of the famous “Rabinowitsch trick” in both the first and third steps. Throughout, E shall be assumed to be a finitely generated congruence on $\mathbf{T}[\mathbf{x}]$.

5.2. The tropically vanishing case

In this section we determine what can be said about pairs of the form $(p, 0_{\mathbf{T}})$ in $\mathbf{E}(\mathbf{V}(E))$. (We are now denoting our polynomial by p to avoid notational confusion with the generators for E .) We pursue a strategy analogous to the standard proof of the classical Nullstellensatz, attempting to “invert” the polynomial p by adjoining the pair $(1_{\mathbf{T}}, yp)$ to the congruence E , working now on the semiring $\mathbf{T}[\mathbf{x}, y]$. Under the assumption that p vanishes identically on $\mathbf{V}(E)$, the resulting congruence F on $\mathbf{T}[\mathbf{x}, y]$ has empty congruence variety, and so Theorem 2 implies $(1_{\mathbf{T}}, 0_{\mathbf{T}}) \in \text{rad}^-(F)$. After “clearing denominators” we will arrive at our result. Most of the actual work is contained in proving the following proposition, which describes the pairs on the original ring $\mathbf{T}[\mathbf{x}]$ that appear in F :

Proposition 5.10. *Let E be a finitely generated congruence on $\mathbf{T}[\mathbf{x}]$, $p \in \mathbf{T}[\mathbf{x}]$ be a nonzero polynomial, and $F = \langle E, (1_{\mathbf{T}}, yp) \rangle_{\mathbf{T}[\mathbf{x}, y]}$. For $a, b \in \mathbf{T}[\mathbf{x}]$, we have $(a, b) \in F$ if and only if $(ap^N, bp^N) \in E$ for some $N > 0$.*

Proof. Fix $a, b \in \mathbf{T}[\mathbf{x}]$. First suppose $(ap^N, bp^N) \in E \subseteq F$ for some $N > 0$. Multiplying by $(y^N, y^N) \in F$, we then obtain $(a(yp)^N, b(yp)^N) \in F$. As $(1_{\mathbf{T}}, yp) \in F$, we thus have $(a, b) \in F$.

Conversely, suppose $(a, b) \in F$. Fix a choice of generators for E , say $(f_1, g_1), \dots, (f_k, g_k)$. Then $F = \langle (f_1, g_1), \dots, (f_k, g_k), (1_{\mathbf{T}}, yp) \rangle_{\mathbf{T}[\mathbf{x}, y]}$, so by Corollary 2.12 there exist $k_i \in \mathbf{T}[\mathbf{x}, y]$ such that

$$(a, k_1), (k_1, k_2), \dots, (k_l, b) \in \mathcal{T},$$

where $\mathcal{T}$ is the collection of all pairs of the form

$$\left(\sum_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \mathbf{f}^{\tilde{\mathbf{m}}} \mathbf{g}^{\tilde{\mathbf{n}}} 1_{\mathbf{T}}^{m'}(yp)^{n'}, \sum_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \mathbf{f}^{\tilde{\mathbf{m}}} \mathbf{g}^{\tilde{\mathbf{n}}} 1_{\mathbf{T}}^{n'}(yp)^{m'} \right),$$

where $s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \in \mathbf{T}[\mathbf{x}, y]$, all but finitely many zero. (In our notation here $\tilde{\mathbf{m}} = (\mathbf{m}, m')$, $\tilde{\mathbf{n}} = (\mathbf{n}, n') \in \mathbf{Z}_{\geq 0}^{k+1}$.) As $E = \langle (f_1, g_1), \dots, (f_k, g_k) \rangle_{\mathbf{T}[\mathbf{x}]}$, similarly let $\mathcal{S}$ denote the collection of all pairs of the form

$$\left(\sum_{\mathbf{m}, \mathbf{n}} r_{\mathbf{m}, \mathbf{n}} \mathbf{f}^{\mathbf{m}} \mathbf{g}^{\mathbf{n}}, \sum_{\mathbf{m}, \mathbf{n}} r_{\mathbf{m}, \mathbf{n}} \mathbf{f}^{\mathbf{n}} \mathbf{g}^{\mathbf{m}} \right),$$

where $r_{\mathbf{m}, \mathbf{n}} \in \mathbf{T}[\mathbf{x}]$, all but finitely many zero. We will show that for N sufficiently large there exist $h_i \in \mathbf{T}[\mathbf{x}]$ such that

$$(ap^N, h_1), (h_1, h_2), \dots, (h_l, bp^N) \in \mathcal{S}.$$

By [Corollary 2.12](#) this will prove $(ap^N, bp^N) \in E$, as desired. For notational convenience, write $a = k_0$ and $b = k_{l+1}$. Now, for each $0 \leq i \leq l$, since $(k_i, k_{i+1}) \in \mathcal{T}$ there exist ${}^i s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \in \mathbf{T}[\mathbf{x}, y]$ such that

$$k_i = \sum_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} {}^i s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \mathbf{f}^{\tilde{\mathbf{m}}} \mathbf{g}^{\tilde{\mathbf{n}}} 1_{\mathbf{T}}^{m'}(yp)^{n'} \quad \text{and} \quad k_{i+1} = \sum_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} {}^i s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \mathbf{f}^{\tilde{\mathbf{n}}} \mathbf{g}^{\tilde{\mathbf{m}}} 1_{\mathbf{T}}^{n'}(yp)^{m'}.$$

Let us write each of these expressions as polynomials in y with coefficients in $\mathbf{T}[\mathbf{x}]$. First write ${}^i s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} = \sum_d {}^i_d s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} y^d$, where ${}^i_d s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \in \mathbf{T}[\mathbf{x}]$. Then we can rewrite the above expressions as

$$k_i = \sum_e \left(\sum_{\substack{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}, d \\ d+n'=e}} {}^i_d s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \mathbf{f}^{\tilde{\mathbf{m}}} \mathbf{g}^{\tilde{\mathbf{n}}} p^{n'} \right) y^e \quad \text{and} \quad k_{i+1} = \sum_e \left(\sum_{\substack{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}, d \\ d+m'=e}} {}^i_d s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \mathbf{f}^{\tilde{\mathbf{n}}} \mathbf{g}^{\tilde{\mathbf{m}}} p^{m'} \right) y^e.$$

For each $1 \leq i \leq l$, the term k_i appears in the pairs (k_{i-1}, k_i) and (k_i, k_{i+1}) . It follows that the two expressions of k_i (as polynomials in y) must be equal, and hence for every e we have

$$\sum_{\substack{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}, d \\ d+m'=e}} {}^{i-1}_d s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \mathbf{f}^{\tilde{\mathbf{n}}} \mathbf{g}^{\tilde{\mathbf{m}}} p^{m'} = \sum_{\substack{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}, d \\ d+n'=e}} {}^i_d s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \mathbf{f}^{\tilde{\mathbf{m}}} \mathbf{g}^{\tilde{\mathbf{n}}} p^{n'}.$$

(The expression on the left comes from considering k_i in the pair (k_{i-1}, k_i) , while the expression on the right comes from considering k_i in the pair (k_i, k_{i+1}) .) If we multiply both sides of the above by p^{N-e} this gives

$$\sum_{\substack{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}, d \\ d+m'=e}} i_d^{-1} s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \mathbf{f}^{\mathbf{n}} \mathbf{g}^{\mathbf{m}} p^{N-d} = \sum_{\substack{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}, d \\ d+n'=e}} i_d s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \mathbf{f}^{\mathbf{m}} \mathbf{g}^{\mathbf{n}} p^{N-d}.$$

Summing these equalities over e yields

$$\sum_{\mathbf{m}, \mathbf{n}} \left(\sum_{m', n', d} i_d^{-1} s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} p^{N-d} \right) \mathbf{f}^{\mathbf{n}} \mathbf{g}^{\mathbf{m}} = \sum_{\mathbf{m}, \mathbf{n}} \left(\sum_{m', n', d} i_d s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} p^{N-d} \right) \mathbf{f}^{\mathbf{m}} \mathbf{g}^{\mathbf{n}}.$$

Let ${}^i r_{\mathbf{m}, \mathbf{n}} = \sum_{m', n', d} i_d s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} p^{N-d}$ and $h_i = \sum_{\mathbf{m}, \mathbf{n}} {}^i r_{\mathbf{m}, \mathbf{n}} \mathbf{f}^{\mathbf{m}} \mathbf{g}^{\mathbf{n}} = \sum_{\mathbf{m}, \mathbf{n}} i_d^{-1} r_{\mathbf{m}, \mathbf{n}} \mathbf{f}^{\mathbf{n}} \mathbf{g}^{\mathbf{m}}$. We then have a transitive chain in $\mathcal{S}$:

$$(h_0, h_1), (h_1, h_2), \dots, (h_l, h_{l+1}) \in \mathcal{S}.$$

It follows that $(h_0, h_{l+1}) \in E$. All that remains is to prove $h_0 = ap^N$ and $h_{l+1} = bp^N$. Setting $i = 0$ in our earlier expression for k_i gives

$$a = \sum_e \left(\sum_{\substack{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}, d \\ d+n'=e}} {}^0 s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \mathbf{f}^{\mathbf{m}} \mathbf{g}^{\mathbf{n}} p^{n'} \right) y^e.$$

Let $\mathcal{C}$ denote the collection of all pairs $(\mathbf{m}, \mathbf{n})$ such that $\mathbf{f}^{\mathbf{m}} \mathbf{g}^{\mathbf{n}} = 0_{\mathbf{T}}$. Since $a \in \mathbf{T}[\mathbf{x}]$, the above equality implies $s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} = 0_{\mathbf{T}}$ whenever $(\mathbf{m}, \mathbf{n}) \notin \mathcal{C}$ and $n' > 0$, and $s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \in \mathbf{T}[\mathbf{x}]$ whenever $(\mathbf{m}, \mathbf{n}) \notin \mathcal{C}$ and $n' = 0$. In particular,

$$a = \sum_{(\mathbf{m}, \mathbf{n}) \notin \mathcal{C}} \left(\sum_{m'} {}^0 s_{\tilde{\mathbf{m}}, (\mathbf{n}, 0)} \right) \mathbf{f}^{\mathbf{m}} \mathbf{g}^{\mathbf{n}}.$$

On the other hand, observe that for every e we have

$$\sum_{(\mathbf{m}, \mathbf{n}) \notin \mathcal{C}} \left(\sum_{\substack{m', n', d \\ m'+d=e}} {}^0 s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} p^{m'} \right) \mathbf{f}^{\mathbf{m}} \mathbf{g}^{\mathbf{n}} = \sum_{(\mathbf{m}, \mathbf{n}) \notin \mathcal{C}} {}^0 s_{(\mathbf{m}, e), (\mathbf{n}, 0)} p^e \mathbf{f}^{\mathbf{m}} \mathbf{g}^{\mathbf{n}}.$$

If we multiply both sides by p^{N-e} and sum over e this gives

$$\sum_{(\mathbf{m}, \mathbf{n}) \notin \mathcal{C}} \left(\sum_{m', n', d} {}^0 s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} p^{N-d} \right) \mathbf{f}^{\mathbf{m}} \mathbf{g}^{\mathbf{n}} = \sum_{(\mathbf{m}, \mathbf{n}) \notin \mathcal{C}} \left(\sum_e {}^0 s_{(\mathbf{m}, e), (\mathbf{n}, 0)} p^N \right) \mathbf{f}^{\mathbf{m}} \mathbf{g}^{\mathbf{n}}.$$

The left-hand side above is precisely h_0 (the missing terms in the sum with $(\mathbf{m}, \mathbf{n}) \in \mathcal{C}$ all vanish), while the right-hand side above is ap^N . We've thus proven $h_0 = ap^N$. The same argument, mutatis mutandis, proves $h_{l+1} = bp^N$. $\square$

We can now prove the following special case of the strong form of the Tropical Nullstellensatz for Congruences:

Theorem 4. *Suppose E is a finitely generated congruence on $\mathbf{T}[\mathbf{x}]$ and $p \in \mathbf{T}[\mathbf{x}]$. Then $(p, 0_{\mathbf{T}}) \in \mathbf{E}(\mathbf{V}(E))$ if and only if $(p, 0_{\mathbf{T}}) \in \text{rad}^-(E)$.*

Proof. One direction is clear, as $\text{rad}^-(E) \subseteq \mathbf{E}(\mathbf{V}(E))$, so suppose $(p, 0_{\mathbf{T}}) \in \mathbf{E}(\mathbf{V}(E))$. Without loss of generality, we may assume p is nonzero. Define $F = \langle E, (1_{\mathbf{T}}, yp) \rangle_{\mathbf{T}[\mathbf{x}, y]}$. Then $\mathbf{V}_{\mathbf{T}^{n+1}}(F) = \emptyset$, so by Theorem 3 it follows that there exists $r \in \mathbf{T}[\mathbf{x}, y]$ such that $(1_{\mathbf{T}} + r, \epsilon(1_{\mathbf{T}} + r)) \in F$ for some $\epsilon \neq 1_{\mathbf{T}}$ (and hence all $\epsilon \in \mathbf{R}$). Write $r(\mathbf{x}, y) = \sum_{0 \leq d \leq M} a_d(\mathbf{x})y^d$. Then $p^M r = \sum_{0 \leq d \leq M} (a_d p^{M-d})(yp)^d$, and so

$$\left(p^M r, \sum_{0 \leq d \leq M} (a_d p^{M-d}) \right) \in F.$$

Let $r' = \sum_{0 \leq d \leq M} a_d p^{M-d} \in \mathbf{T}[\mathbf{x}]$. Then by the above observation we have $(1_{\mathbf{T}} + r', \epsilon(1_{\mathbf{T}} + r')) \in F$. By Proposition 5.10, it follows that $(p^N + r'', \epsilon(p^N + r'')) \in E$ for some $N \geq 1$ (where $r'' = p^N r' \in \mathbf{T}[\mathbf{x}]$). This proves $(p, 0_{\mathbf{T}}) \in \text{rad}^-(E)$. $\square$

Corollary 5.11. *Suppose E is a flat, finitely generated congruence on $\mathbf{T}[\mathbf{x}]$, and $p \in \mathbf{T}[\mathbf{x}]$. Then $(p, 0_{\mathbf{T}}) \in \mathbf{E}(\mathbf{V}(E))$ if and only if $(p^N, 0_{\mathbf{T}}) \in E$ for some $N > 0$.*

Proof. One direction is obvious, so suppose $(p, 0_{\mathbf{T}}) \in \mathbf{E}(\mathbf{V}(E))$. By Theorem 4, we have $(p^N + A, \epsilon(p^N + A)) \in E$ for some $A \in \mathbf{T}[\mathbf{x}]$, $N > 0$ and $\epsilon \neq 1_{\mathbf{T}}$. As E is flat, this implies $(p^N + A, 0_{\mathbf{T}}) \in E$, and hence $(p^N, 0_{\mathbf{T}}) \in E$. $\square$

5.3. The tropically constant case

In this section we will determine what can be said when a polynomial is constant (and nonzero) on $\mathbf{V}(E)$. It suffices to consider the case when $(p, 1_{\mathbf{T}}) \in \mathbf{E}(\mathbf{V}(E))$. Note that in this case we have $p(\mathbf{a}) < t$ for every $t > 1_{\mathbf{T}}$, and so $(p + t, t) \in \mathbf{E}(\mathbf{V}(E))$ for every $t > 1_{\mathbf{T}}$. Similarly, $(p + s, p) \in \mathbf{E}(\mathbf{V}(E))$ for every $s < 1_{\mathbf{T}}$. We are thus in the situation of Example 5.7. Our main work will be in proving the following:

Proposition 5.12. *Suppose E is a finitely generated congruence on $\mathbf{T}[\mathbf{x}]$, and suppose $p, q \in \mathbf{T}[\mathbf{x}]$ are such that $p(\mathbf{a}) < q(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{V}(E)$. Then $(p + q, q) \in \text{rad}(E)$.*

The proof of this proposition will rely on the following general lemma (which holds in any additively idempotent semiring):

Lemma 5.13. *Let R be an additively idempotent semiring, and let m, n be fixed nonnegative integers. For sufficiently large N , and any $a, b \in R$, we have the following equality:*

$$(a+b)^N(a+b)^ma^n = a^N(a+b)^na^m + b(a+b)^{N-1}(a+b)^ma^n.$$

Proof. First suppose $m \leq n$. Observe that

$$\begin{aligned} (a+b)^N(a+b)^ma^n &= (a+b)^{N-(n-m)}a^{n-m}(a+b)^na^m \\ &= \left(a^N + a^{N-1}b + \dots + a^{n-m}b^{N-(n-m)}\right)(a+b)^na^m \\ &= a^N(a+b)^na^m + \left(a^{N-1}b + \dots + a^{n-m}b^{N-(n-m)}\right)(a+b)^na^m. \end{aligned}$$

Notice that the second term on the right-hand side of the above equation equals

$$\begin{aligned} b\left(a^{N-1} + \dots + a^{n-m}b^{N-1-(n-m)}\right)(a+b)^na^m &= b\left(a^{N-1+n} + \dots + a^{n-m}b^{N-1+m}\right)a^m \\ &= b(a+b)^{N-1+m}a^{n-m}a^m \\ &= b(a+b)^{N-1}(a+b)^ma^n. \end{aligned}$$

This proves the statement in the case $m \leq n$. The case when $m > n$ is proved similarly. In that case, observe that since $(a+b)^N = a^N + b(a+b)^{N-1}$, we have

$$(a+b)^N(a+b)^ma^n = a^N(a+b)^ma^n + b(a+b)^{N-1}(a+b)^ma^n. \quad (*)$$

Notice that the first term on the right-hand side of the above equation equals

$$\begin{aligned} a^N(a+b)^ma^n &= a^{N-(m-n)}(a+b)^{m-n}(a+b)^na^m \\ &= \left(a^N + a^{N-1}b + \dots + a^{N-(m-n)}b^{m-n}\right)(a+b)^na^m \\ &= a^N(a+b)^na^m + b\left(a^{N-1} + \dots + a^{N-(m-n)}b^{m-n-1}\right)(a+b)^na^m. \end{aligned}$$

We claim the second term on the right-hand side of the above equation is a subsum of the second term on the right-hand side of (*). Indeed, observe first that

$$\begin{aligned} b\left(a^{N-1} + \dots + a^{N-(m-n)}b^{m-n-1}\right)(a+b)^na^m \\ = b\left(a^{N-1+(m-n)} + \dots + a^Nb^{m-n-1}\right)(a+b)^na^n. \end{aligned}$$

Then also observe that

$$\begin{aligned} b(a+b)^{N-1}(a+b)^m a^n &= b(a+b)^{N-1+(m-n)}(a+b)^n a^n \\ &= b \left(a^{N-1+(m-n)} + \dots + b^{N-1+(m-n)} \right) (a+b)^n a^n. \end{aligned}$$

Thus, the claim also holds in the case $m > n$. $\square$

Proof of Proposition 5.12. Fix generators $(f_1, g_1), \dots, (f_k, g_k)$ for E . Let $F = \langle E, (p+q, p) \rangle$. Then $\mathbf{V}(F) = \emptyset$, so by Theorem 3 there exists $r \in \mathbf{T}[\mathbf{x}]$ such that $(1_{\mathbf{T}} + r, \epsilon(1_{\mathbf{T}} + r)) \in F$ for every $\epsilon \in \mathbf{R}$. Fix any $\epsilon \in \mathbf{R}$. By Corollary 2.12 there exist $k_i \in \mathbf{T}[\mathbf{x}]$ such that

$$(1_{\mathbf{T}} + r, k_1), (k_1, k_2), \dots, (k_l, \epsilon(1_{\mathbf{T}} + r)) \in \mathcal{T},$$

where $\mathcal{T}$ is the collection of all pairs of the form

$$\left(\sum_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \mathbf{f}^{\tilde{\mathbf{m}}} \mathbf{g}^{\tilde{\mathbf{n}}} (p+q)^{m'} p^{n'}, \sum_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \mathbf{f}^{\tilde{\mathbf{m}}} \mathbf{g}^{\tilde{\mathbf{m}}} (p+q)^{n'} p^{m'} \right),$$

where $s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \in \mathbf{T}[\mathbf{x}]$, all but finitely many zero. (In our notation here $\tilde{\mathbf{m}} = (\mathbf{m}, m')$, $\tilde{\mathbf{n}} = (\mathbf{n}, n') \in \mathbf{Z}_{\geq 0}^{k+1}$.) For notational convenience, write $k_0 = 1_{\mathbf{T}} + r$ and $k_{l+1} = \epsilon(1_{\mathbf{T}} + r)$. Now, for each $0 \leq i \leq l$, since $(k_i, k_{i+1}) \in \mathcal{T}$ there exist ${}^i s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \in \mathbf{T}[\mathbf{x}]$ such that

$$k_i = \sum_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} {}^i s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \mathbf{f}^{\tilde{\mathbf{m}}} \mathbf{g}^{\tilde{\mathbf{n}}} (p+q)^{m'} p^{n'} \quad \text{and} \quad k_{i+1} = \sum_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} {}^i s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \mathbf{f}^{\tilde{\mathbf{n}}} \mathbf{g}^{\tilde{\mathbf{m}}} (p+q)^{n'} p^{m'}.$$

Observe that by Lemma 5.13

$$\begin{aligned} (p+q)^N k_i &= \sum_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} {}^i s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \mathbf{f}^{\tilde{\mathbf{m}}} \mathbf{g}^{\tilde{\mathbf{n}}} (p+q)^N (p+q)^{m'} p^{n'} \\ &= p^N \left(\sum_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} {}^i s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \mathbf{f}^{\tilde{\mathbf{m}}} \mathbf{g}^{\tilde{\mathbf{n}}} (p+q)^{n'} p^{m'} \right) \\ &\quad + q(p+q)^{N-1} \left(\sum_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} {}^i s_{\tilde{\mathbf{m}}, \tilde{\mathbf{n}}} \mathbf{f}^{\tilde{\mathbf{m}}} \mathbf{g}^{\tilde{\mathbf{n}}} (p+q)^{m'} p^{n'} \right) \\ &\sim_E p^N k_{i+1} + q(p+q)^{N-1} k_i. \end{aligned}$$

If we now add $q(p+q)^{N-1} k_{i+1}$ to both sides of this equivalence, we obtain

$$(p+q)^N k_i + q(p+q)^{N-1} k_{i+1} \sim_E (p+q)^N k_{i+1} + q(p+q)^{N-1} k_i,$$

or in other words

$$(k_i(p+q)^{N-1}, k_{i+1}(p+g)^{N-1}) \rtimes (p+q, q) \in E.$$

Since $k_0(p+q)^{N-1} = (p+q)^{N-1} + r'$ and $k_{l+1} = \epsilon k_0$, this proves $(p+q, q) \in \text{rad}(E)$. $\square$

Corollary 5.14. *Suppose E is a finitely generated congruence on $\mathbf{T}[\mathbf{x}]$. Then $(p, 1_{\mathbf{T}}) \in \mathbf{E}(\mathbf{V}(E))$ if and only if for every $t > 1_{\mathbf{T}}$ we have $(p+t, t) \in \text{rad}(E)$ and $(p+t^{-1}, p) \in \text{rad}(E)$.*

Proof. If $(p+t, t) \in \text{rad}(E) \subseteq \mathbf{E}(\mathbf{V}(E))$, then $p(\mathbf{a}) \leq t$ for every $\mathbf{a} \in \mathbf{V}(E)$. If this holds for every $t > 1_{\mathbf{T}}$, then we must actually have $p(\mathbf{a}) \leq 1_{\mathbf{T}}$ for every $\mathbf{a} \in \mathbf{V}(E)$, i.e., $(p+1_{\mathbf{T}}, 1_{\mathbf{T}}) \in \mathbf{E}(\mathbf{V}(E))$. Similarly, if $(p+t^{-1}, p) \in \text{rad}(E) \subseteq \mathbf{E}(\mathbf{V}(E))$ for every $t > 1_{\mathbf{T}}$, then we must actually have $(p+1_{\mathbf{T}}, p) \in \mathbf{E}(\mathbf{V}(E))$. It then follows by transitivity that $(p, 1_{\mathbf{T}}) \in \mathbf{E}(\mathbf{V}(E))$.

Conversely, suppose $(p, 1_{\mathbf{T}}) \in \mathbf{E}(\mathbf{V}(E))$, and so $p(\mathbf{a}) = 1_{\mathbf{T}}$ for every $\mathbf{a} \in \mathbf{V}(E)$. Take any $t > 1_{\mathbf{T}}$. Then $p(\mathbf{a}) < t$ for every $\mathbf{a} \in \mathbf{V}(E)$, so by [Proposition 5.12](#) we have $(p+t, t) \in \text{rad}(E)$. Similarly, we have $t^{-1} < p(\mathbf{a})$ for every $\mathbf{a} \in \mathbf{V}(E)$, so [Proposition 5.12](#) also gives $(p+t^{-1}, p) \in \text{rad}(E)$. $\square$

5.4. The general case

At last we consider the general case, and prove the following strong form of the Tropical Nullstellensatz for Congruences. The proof will involve a second use of the Rabinowitsch trick, this time to reduce the general case to the tropically constant case.

Theorem 5. *Suppose E is a finitely generated congruence on $\mathbf{T}[\mathbf{x}]$. Then $\mathbf{E}(\mathbf{V}(E)) = \hat{E}$.*

Proof. We have already seen $\hat{E} \subseteq \mathbf{E}(\mathbf{V}(E))$, so it only remains to prove the reverse containment. Take any $(p, q) \in \mathbf{E}(\mathbf{V}(E))$. If $(p, 0_{\mathbf{T}}) \in \mathbf{E}(\mathbf{V}(E))$, then we also have $(q, 0_{\mathbf{T}}) \in \mathbf{E}(\mathbf{V}(E))$, and by [Theorem 4](#) it then follows that $(p, 0_{\mathbf{T}}), (q, 0_{\mathbf{T}}) \in \text{rad}(E) \subseteq \hat{E}$. Thus $(p, q) \in \hat{E}$.

Now suppose $(p, 0_{\mathbf{T}}) \notin \mathbf{E}(\mathbf{V}(E))$, and so also $(q, 0_{\mathbf{T}}) \notin \mathbf{E}(\mathbf{V}(E))$. Define $F = \langle E, (py, 1_{\mathbf{T}}) \rangle_{\mathbf{T}[\mathbf{x}, y]}$. Then $(qy, 1_{\mathbf{T}}) \in \mathbf{E}(\mathbf{V}(F))$, and so by [Corollary 5.14](#) it follows that for every $t > 1_{\mathbf{T}}$ we have $(qy+t, t) \in \text{rad}(F)$. This implies there exist $h_0, \dots, h_{l+1} \in \mathbf{T}[\mathbf{x}, y]$ such that for every i we have

$$(h_i, h_{i+1}) \rtimes (qy+t, t) \in F,$$

where $h_0 = (qy+t)^N + r$ and $h_{l+1} = \epsilon h_0$ for some $r \in \mathbf{T}[\mathbf{x}, y], \epsilon \neq 1_{\mathbf{T}}$ and $N > 0$. Multiplying all of these pairs by p^{M+1} and using the fact that $(py, 1_{\mathbf{T}}) \in F$, it follows that we have

$$(h'_i, h'_{i+1}) \rtimes (q+tp, tp) \in F$$

for every i , where $h'_i \in \mathbf{T}[\mathbf{x}]$, $h_0 = p^M(q + tp)^N + r'$, $h_{l+1} = \epsilon h'_0$, and $r' \in \mathbf{T}[\mathbf{x}]$. By [Proposition 5.10](#), it follows that if we multiply all of these pairs once more by $p^{M'}$ we have

$$(h''_i, h''_{i+1}) \rtimes (q + tp, tp) \in E, \quad (1)$$

where $h''_i \in \mathbf{T}[\mathbf{x}]$, $h''_0 = p^{M+M'}(q + tp)^N + r''$, and $h''_{l+1} = \epsilon h''_0$. Multiplying by $p^N(q + tp)^{M+M'}$ if necessary, we may assume $M + M' = N$. Multiplying by t^N , we may further assume $h''_0 = (tp)^N(q + tp)^N + r''$. [Corollary 5.14](#) also implies that for every $t > 1_{\mathbf{T}}$ we have $(qy + t^{-1}, qy) \in \text{rad}(F)$, or equivalently $(tqy + 1_{\mathbf{T}}, tqy) \in \text{rad}(F)$. Repeating the above argument, we find that we have

$$(k_i, k_{i+1}) \rtimes (tq + p, tq) \in E \quad (2)$$

for some $k_i \in \mathbf{T}[\mathbf{x}]$, where $k_0 = p^N(tq + p)^N + s$ for some $s \in \mathbf{T}[\mathbf{x}]$, and $k_{l+1} = \epsilon k_0$.

If we now repeat this entire argument from the beginning with the roles of p and q reversed, we will similarly find that

$$(H_i, H_{i+1}) \rtimes (p + tq, tq) \in E \quad (3)$$

for some $H_i \in \mathbf{T}[\mathbf{x}]$, where $H_0 = (tq)^N(p + tq)^N + R$ for some $R \in \mathbf{T}[\mathbf{x}]$, and $H_{l+1} = \epsilon H_0$; and also that

$$(K_i, K_{i+1}) \rtimes (tp + q, tp) \in E \quad (4)$$

for some $K_i \in \mathbf{T}[\mathbf{x}]$, where $K_0 = q^N(tp + q)^N + S$ for some $S \in \mathbf{T}[\mathbf{x}]$, and $K_{l+1} = \epsilon K_0$. Now observe that if we add (1) and (4) and note that $((tp)^N + q^N)(q + (tp))^N = (q + tp)^{2N}$, it follows that $(q + tp, tp) \in \text{rad}(E)$. We then have an ascending sequence $\{(q + tp, tp)\}_{t > 1_{\mathbf{T}}}$ in $\text{rad}(E)$, from which it follows that $(q + p, p) \in (\text{rad}(E))^{\text{lim}}$. Similarly, summing (2) and (3) produces the ascending sequence $\{(tq + p, tq)\}_{t > 1_{\mathbf{T}}}$ in $\text{rad}(E)$, and hence $(q + p, q) \in (\text{rad}(E))^{\text{lim}}$. Thus, $(p, q) \in \hat{E}$. $\square$

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